MOG JEE Main Full Test - 2024 Revised

Total Time: 180 minutes | Total Marks: 300 | Total Questions: 75

GENERAL INSTRUCTIONS

The Question Paper consists of three parts (Physics, Chemistry and Mathematics) with a total of **75** questions.
Each subject is divided into 2 sections:
Section 1: Single Correct Type Questions (4 options, only one correct)
Section 2: Numerical Value Type Questions (answer to be entered as numerical value)
Marking Scheme:
Single Correct: +4 for correct, -1 for wrong, 0 if unattempted.
Numerical Value: +4 for correct, -1 for wrong.

By: C I P H Ξ R

Q.1

The respective speeds of the five molecules are 1,2,3,4 and $5\mathrm{kms}^{-1}$. Then, the ratio of their rms speed and the average speed will be

(A) $\sqrt{11} : 3$

(B) $3 : \sqrt{11}$

(C) $1 : 2$

(D) $3 : 4$

Q.2

A solid iron ball and an aluminium ball of same diameter are released together in a lake. Which ball will reach the bottom first. (neglect viscosity and $\rho_{\mathrm{Al}} < \rho_{\mathrm{Fe}}$)

(A) Aluminium Ball

(B) Iron Ball

(C) Both will reach at the same time

(D) Data insufficient

Q.3

A light of wavelength $4000\tilde{A}\ldots$ is incident on a metal having threshold frequency $5.5 \times 10^{14}\ \mathrm{Hz}$. If light has intensity $\mathrm{I} = 100\ \mathrm{W/m^2}$ then the number of incident photons per m^2 per second, is :-

(A) 1×10^{20}

(B) 2×10^{20}

(C) 3×10^{20}

(D) 4×10^{20}

Q.4

A student measures the time period of 20 oscillation of a simple pendulum five times. That data set is $53\ \text{s}, 52\ \text{s}, 55\ \text{s}, 54\ \text{s}$ and $51\ \text{s}$. If the minimum division in the measuring clock is $1\ \text{s}$, then the reported time should be

(A) $50 \pm 1\text{sec}$

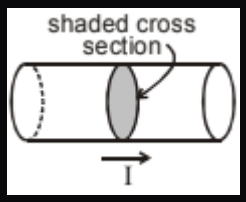
(B) $52 \pm 2\text{sec}$

(C) $53 \pm 2\text{sec}$

(D) $53 \pm 3\text{sec}$

Q.5

A current I flows through a cylindrical rod of uniform cross-section area A and resistivity ρ. The electric flux through the shaded cross-section of rod as shown in figure is:



(A) $\frac{I}{\rho}$

(B) $\rho \mathrm{I}$

(C) $\frac{\rho I}{A}$

(D) $\frac{\rho \mathrm{A}}{\mathrm{I}}$

Q.6

Statement (I) : The temperature coefficient of resistance for most of metals in pure form is positive.
Statement (II) : A metal wire $2\ \text{mm}$ in diameter carries a charge of $360\pi \text{C}$ in two hours. If the metal contains 5×10^{22} free electrons $/\text{cm}^3$, then drift velocity of the electrons in the wire is $6.25 \times 10^{-6}\ \text{m/s}$.
Statement (III) : Semiconductors like pure germanium does not obey Ohm's law for all range of electric field values.
Which of the following is correct?

(A) Statements I, II, III are true.

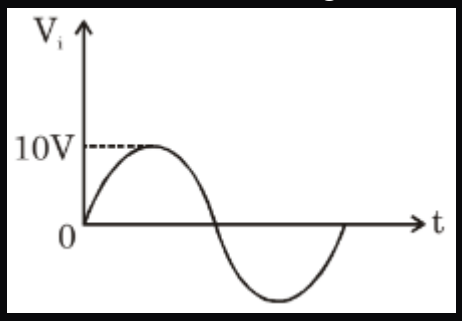
(B) Statements I, II are true, but statement III is false.

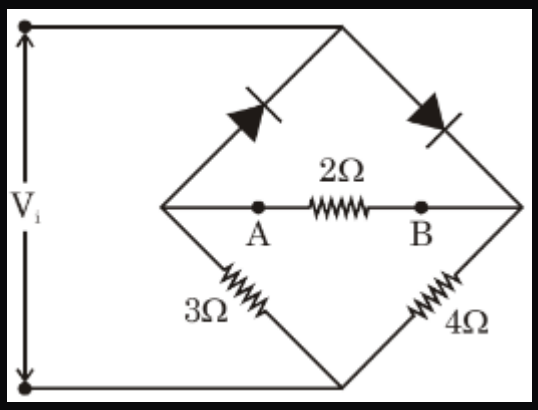
(C) Statements II, III are true, but statement I is false.

(D) Statements I, II, III are false.

Q.7

In the circuit shown in figure, find the magnitude of maximum potential difference (in V) between A & B ?





(A) 1.5

(B) 2

(C) 4

(D) 0.5

Q.8

A galvanometer, having a resistance of 50Ω, gives a full scale deflection for a current of 0.05 A. The length in metre of a resistance wire of area of cross-section $2.97 \times 10^{-2}\ \mathrm{cm}^2$ that can be used to convert the galvanometer into an ammeter which can read a maximum of 5 A current is :
(Specific resistance of the wire $= 5 \times 10^{-7}\Omega - \mathrm{m}$)

(A) 9

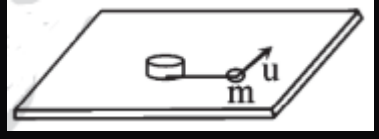
(B) 6

(C) 3

(D) 1.5

Q.9

A particle of mass m is attached to a long ideal string. The other end of the string is joined to a vertical cylinder fixed on a frictionless horizontal plane as shown in the figure. When the particle is projected with speed u in a direction perpendicular to the string, the particle continues to move on a spiral path of decreasing radius till it strikes the cylinder. While the particle is moving on the spiral path, how will the tension in the string and speed of the particle change?



(A) Both will remains constant

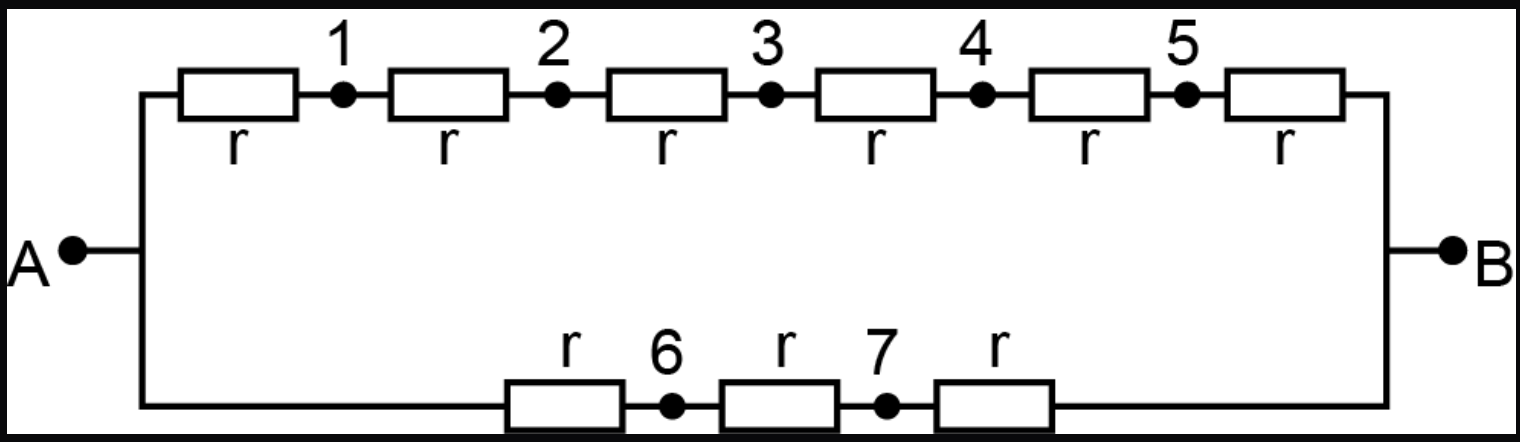
(B) Both will increase

(C) Tension will increase and speed will decrease

(D) Tension will increase and speed will remain constant

Q.10

In the circuit shown in the figure all the resistance are identical and each has the value $r\Omega$. The equivalent resistance of the combination between the points A and B will remain unchanged even when the following pairs of points marked in the figure are connected through a resistance $\mathbb{R}$.



(A) 2 and 6

(B) 3 and 6

(C) 4 and 7

(D) Option (a) and (c)

Q.11

For an EM wave: $\mathrm{E}=\mathrm{E}_0 \sin 10^7\left[\mathrm{x}-2.5 \times 10^8 \mathrm{t}\right] \quad \mathrm{N/C}$ in a medium then its refractive index is :

(A) 1.5

(B) 1.2

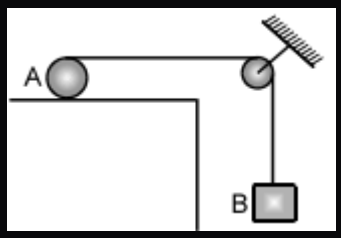
(C) 1.33

(D) 1.66

Q.12

Find the acceleration of solid right circular roller A, weighing $12\ \mathrm{kg}$ when it is being pulled by another weight $\mathrm{B}(6\ \mathrm{kg})$ along the horizontal plane as in figure (pulley is massless). The weight B is attached to the end of a string

wound around the circumference of roller. Assume there is no slipping of the roller and the string is inextensible.



(A) $4\ \mathrm{g}/7$

(B) $\mathrm{g}/7$

(C) $2\ \mathrm{g}/7$

(D) $3\ \mathrm{g}/7$

Q.13

Parallel plate capacitor is connected to a battery through resistance ' R '. At steady state, plates of capacitor are moved away very slowly. Then charge on capacitor plate

(A) Will increase

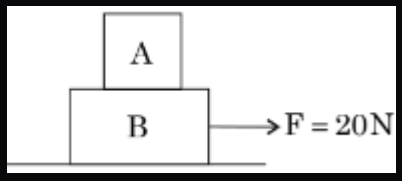
(B) Will remain same

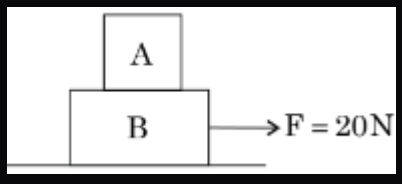
(C) Will decrease

(D) Will depend on polarity of battery

Q.14

Block A of mass $2\ \mathrm{kg}$ is placed over block B of mass $8\ \mathrm{kg}$. The combination is placed over a smooth horizontal surface. Coefficient of friction between A and B is 0.5. A horizontal force of $20\ \mathrm{N}$ is applied on block B. The force of friction on A is





(A) $10\ \mathrm{N}$

(B) $8\ \mathrm{N}$

(C) $6\ \mathrm{N}$

(D) $4\ \mathrm{N}$

Q.15

In a step-up transformer the turns ratio is 10. If the frequency of the current in the primary coil is $50\ \text{Hz}$ then the frequency of the current in the secondary coil will b

(A) $500\ \text{Hz}$

(B) $5\ \text{Hz}$

(C) $60\ \text{Hz}$

(D) $50\ \text{Hz}$

Q.16

The kinetic energy (in keV) of the α-particle, when the nucleus $^{210}_{84}\text{Po}$ (polonium) at rest undergoes α-decay is (apporoximately)
[masses of atoms are given as
$^{4}_{2}\text{He} = 4.002603\text{u}, ^{206}_{82}\ \text{Pb} = 205.974455\text{u},\ ^{10}_{84}\text{Po} = 209.982876\text{u}]$

(A) 5319

(B) 5422

(C) 5707

(D) 5818

Q.17

Consider a vechicle moving with a velocity $54\ \text{km/h}$. At a distance of $400\ \text{m}$, from the traffic light brakes are applied. The acceleration of the vehicle, after the application of brakes is $-0.3\ \text{m/s}^2$. The vehicle's position relative to the traffic light is (till it stops)

(A) $25\ \text{m}$

(B) $375\ \text{m}$

(C) $425\ \text{m}$

(D) $30\ \text{m}$

Q.18

In a single slit diffraction experiment first minima for $\lambda_1 = 660\ \text{nm}$ coincides with the first maxima for wavelength λ_2. Calculate λ_2 (approximately).

(A) $660\ \text{nm}$

(B) $220\ \text{nm}$

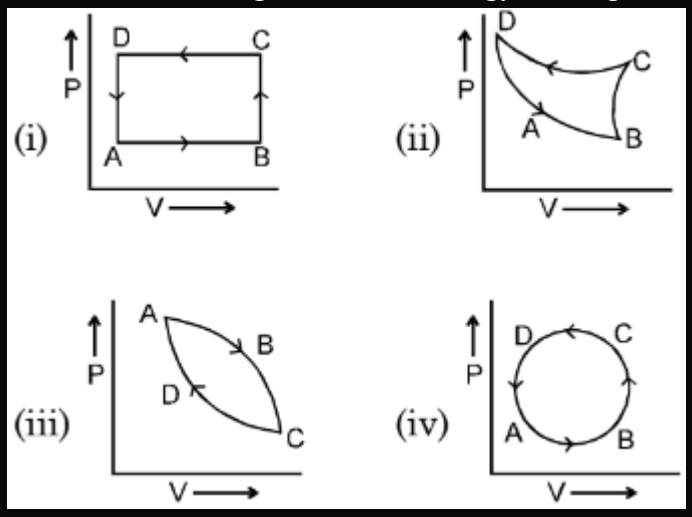
(C) $440\ \text{nm}$

(D) $1320\ \text{nm}$

Q.19

In the following figures (i) to (iv), variation of volume by change of pressure is shown. A gas is taken along the path

ABCDA. The change in internal energy of the gas will be:



(A) positive in all cases from (i) to (iv)

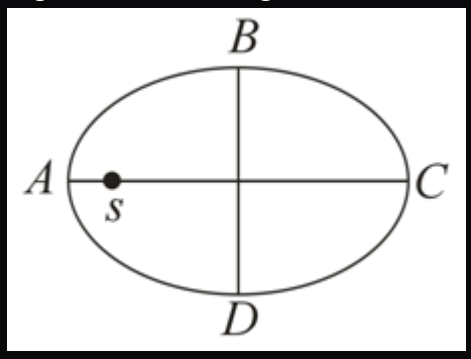
(B) positive in cases (i), (ii) and (iii) but zero in case (iv)

(C) negative in cases (i), (ii) and (iii) but zero in case (iv)

(D) zero in all the four cases.

Q.20

A planet is revolving around the sun in which of the following is correct statement?



(A) The time taken in travelling DAB is less than that for BCD

(B) The time taken in travelling DAB is greater than that for BCD

(C) The time taken in travelling CDA is less than that for ABC

(D) The time taken in travelling CDA is lgreater than that for ABC

Q.21

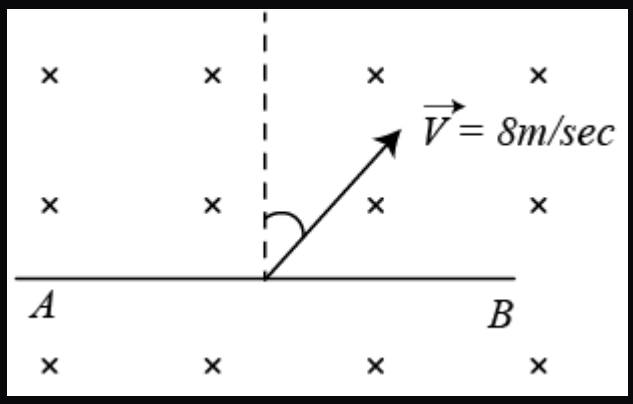
The deflection produced in a tangent galvanometer, whose coil has a resistance of 9Ω is 30°. The potential difference across the coil is $4.5\ \mathrm{V}$. If the number of turns in the coil is 10 , the radius of the coil is x cm, find x^2. (Given, $B_{\mathrm{H}}=3.14 \times 10^{-5}\ \mathrm{T}$)

Q.22

In Young's double slit experiment, red light of wavelength $6000 \mathring{A}$ is used and the nth bright fringe is obtained at a point P on the screen. Keeping the same setting, the source of light is replaced by green light of wavelength $5000 \mathring{A}$ and now $(n+1)$ th bright fringe is obtained at the point P on the screen. The value of n is

Q.23

A metal rod AB of length $50\ \text{cm}$ is moving at a velocity $8\ \text{ms}^{-1}$ in a magnetic field of $2\ \text{T}$. If the field is at 60° with the plane of motion as shown in the figure, then the potentials V_A and V_B are related by $V_A - V_B = K$, find K.



Q.24

In a hydrogen atom, electron moves from second excited state to first excited state and then from first excited state to ground state. Ratio of wavelength is K/10, find K.

Q.25

A point charge $+q$ is fixed at point A on the circumference of a fixed smooth non-conducting circular wire of radius R in a gravity free space. A small ring of mass m and charge $+q$ which can slide freely on the wire track is given tangential velocity at point B which is diametrically opposite to A. If the distance of closest approach between the fixed charge and the ring is R, then force applied by the wire track on the ring just after the release will be xF, find value of $12x$. $\left[F = \frac{1}{4\pi\in_0}\frac{q^2}{R^2}\right]$.

Q.26

Which of the following pairs of ions are colourless?

(A) $\text{Ti}^{3+}, \text{Cu}^{2+}$

(B) $\text{Sc}^{3+}, \text{Zn}^{2+}$

(C) $\text{Co}^{2+}, \text{Fe}^{3+}$

(D) $\text{Ni}^{2+}, \text{V}^{3+}$

Q.27

What volume of $0.2\text{M} - \text{KMnO}_4$ solution is required for complete reaction with $20\text{ml}\ 0.4\text{M} - \text{FeC}_2\text{O}_4$ solution in presence of H_2SO_4 ?

(A) 24ml

(B) 8ml

(C) 16ml

(D) 120ml

Q.28

Identify the set of acidic oxides.

(A) Na_2O, CaO, BaO

(B) ZnO, PbO, BeO

(C) CO, NO, N_2O

(D) $Mn_2O_7, CrO_3,\ V_2O_5$

Q.29

Find incorrect statement for coordination compounds.

(A) $[PtCl_2\,(NH_3)_2]$ & $[Cu\,(NH_3)_4]^{2+}$ are square planar complexes.

(B) Complex $[Fe\,(C_2O_4)_3]^{3-}$, does not exhibit geometrical isomerism while exhibits optical isomerism.

(C) CO bond order : $[Ni(CO)_4] > [Co(CO]_4]^- > [Fe(CO)_4]^{2-}$.

(D) $[Ni\,(Cl_2)\,(PPh_3)_2]$ has two geometrical isomers.

Q.30

Given below are two statements
Assertion (A): The enthalpy of formation of graphite is taken as zero.
Reason (R): Graphite is thermodynamically most stable allotrope of carbon
The correct answer is

(A) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(B) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(C) (A) is correct but (R) is incorrect

(D) (A) is incorrect but (R) is correct

Q.31

Select correct for $O-O$ bond order.

(A) $O_3 > O_2^+ > H_2O_2 > O_2$

(B) $O_2^+ > O_2 > O_3 > H_2O_2$

(C) $O_2^+ > O_2 > H_2O_2 > O_3$

(D) $O_2^+ > O_3 > O_2 > H_2O_2$

Q.32

For which of the following, Kjeldahl's method is not used for the estimation of nitrogen?

I. Aniline

II. Azobenzene

III. Nitrobenzene

IV. Pyridine

(A) II, III, IV

(B) II, III only

(C) III, IV only

(D) I, III, IV

Q.33

The schrodinger wave equation for hydrogen atom is

$$\psi_{\text{radial}} = \frac{1}{16\sqrt{4}}\left(\frac{\mathrm{Z}}{\mathrm{a}_0}\right)^{3/2}\left[(\sigma-1)\left(\sigma^2-8\sigma+12\right)\right]\mathrm{e}^{-\sigma/2}$$

$$\text{where } \sigma = \frac{2\mathrm{Zr}}{\mathrm{a}_0}$$

(a_0 & Z are constants in which the answer can be expressed).

Minimum & maximum distance of radial node from nucleus respectively are

(A) $\frac{a_0}{Z}, \frac{3a_0}{Z}$

(B) $\frac{\mathrm{a}_0}{2\mathrm{Z}}, \frac{\mathrm{a}_0}{\mathrm{Z}}$

(C) $\frac{\mathrm{a}_0}{2\mathrm{Z}}, \frac{3\mathrm{a}_0}{\mathrm{Z}}$

(D) $\frac{\mathrm{a}_0}{2\mathrm{Z}}, \frac{4\mathrm{a}_0}{\mathrm{Z}}$

Q.34

Which of the following reactions produce ethyl cyanide as a major product?

1. $\mathrm{CH_3CH_2CH=NOH} \xrightarrow[\mathrm{K_2CO_3}]{(\mathrm{CH_3O})_2\mathrm{O}}$
2. $\mathrm{CH_3CH_2Cl} \xrightarrow{\text{Ethanolic KCN}}$
3. $\mathrm{CH_3CH_2Cl} \xrightarrow{\text{Ethanolic AgCN}}$
4. $\mathrm{CH_3CH_2CON(CH_3)_2} \xrightarrow[\text{Pyridine, } 70^\circ\mathrm{C}]{\mathrm{C_6H_5SO_2Cl}}$

(A) 1,3

(B) 2,3

(C) 3,4

(D) 1,2

Q.35

Which one of the following coordination complexes exhibit the lowest value of magnetic moment (in BM)?

(A) $[\mathrm{Cr(CN)_6}]^{3-}$

(B) $[Mn(CN)_6]^{3-}$

(C) $[Fe(CN)_6]^{3-}$

(D) $[Co(CN)_6]^{3-}$

Q.36

As per VSEPR theory in which molecular species nitrogen has $NX_3\ L$ type arrangement, but lone pair is NOT localised on nitrogen.
(N = nitrogen, X = σ bonds, L = Lone pair)

(A) $N(CH_3)_3$

(B) $N(CH_3)_4^+$

(C) CH_3NH_2

(D) $CH_3\ N(SiH_3)_2$

Q.37

Which of the following statements are correct?
(i) Inductive effect and resonance effect is possible in chlorobenzene.
(ii) Resonance effect dominates over inductive effect in anisole.
(iii) p-nitrobenzoic acid is less acidic than m-nitrobenzoic acid.
(iv) Diphenylamine is more basic than aniline.

(A) (i), (ii), (iii) and (iv)

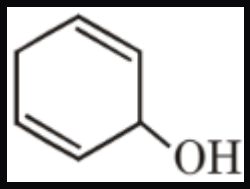
(B) (i) and (ii)

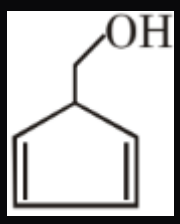
(C) (iii) and (iv)

(D) (i) and (iii)

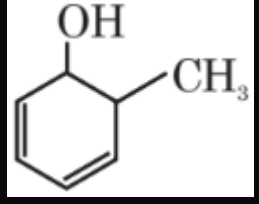
Q.38

Which alcohols on reaction with concentrated H_2SO_4 at high temperature will give product, which give +ve test with Br_2 solution :

(A) 

(B) 

(C) $Me_3C-CH_2-CH_2OH$

(D) 

Q.39

Which of the following characteristics of the transition metals is associated with their catalytic activity?

(A) Colour of hydrated ions

(B) Diamagnetic behaviour

(C) Paramagnetic behaviour

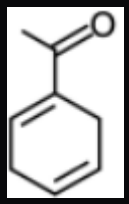
(D) Variable oxidation states

Q.40

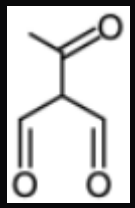
Which of the following option(s) is/are correct for the products of the given reaction sequence?

$$\text{Acetophenone} \xrightarrow{\text{Na in liquid } NH_3} \text{H} \xrightarrow{O_3} \text{I} \xrightarrow{Zn/H_2O} \text{J (contains 5C atoms)} + \text{K}$$

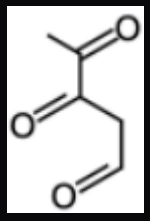
(A) Compound H is



(B) Compound J is



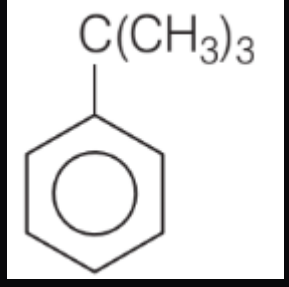
(C) Compound J is

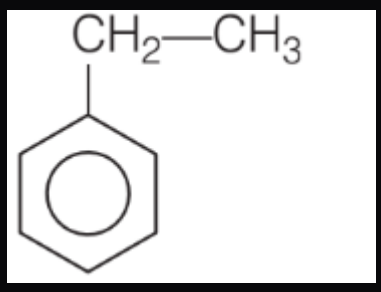


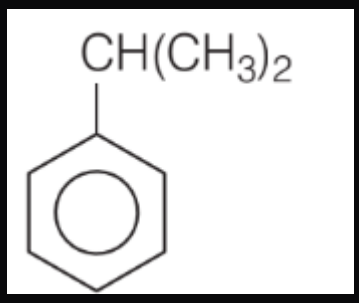
(D) None of these

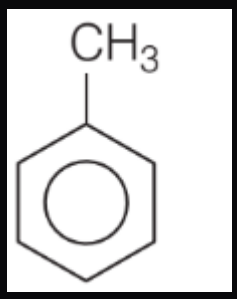
Q.41

Identify the compound, which has maximum number of no bond resonance structures.

(A) 

(B) 

(C) 

(D) 

Q.42

CORRECT statement regarding orthoboric acid is :

(A) It is a Lewis acid whose pKa is -1.

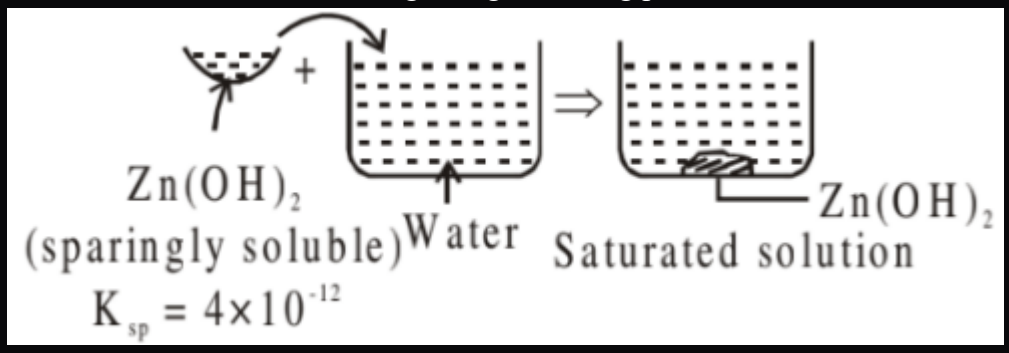
(B) It is a triprotic, weak Bronsted acid.

(C) There exist H-bonds in between layers in its crystalline form.

(D) Cis-diols can be used to increase acidic strength in its aq. solution.

Q.43

Select the correct statements regarding following process.



I. Solubility of $Zn(OH)_2$ is 10^{-4} mol/ lit.
II. On addition of small quantity of OH^- solubility decreases
III. On addition of conc. HCl solubility increases
IV. In presence of $0.01MZnCl_2$, solubility becomes 10^{-5} mol/lit.

(A) I & II

(B) I, II & IV

(C) I & IV

(D) I, II, III & IV

Q.44

The resistance of $0.05M$ solution of oxalic acid is $200ohm$ and cell constant is $2.0\ cm^{-1}$, the equivalent conductance (in $S\ cm^2eq^{-1}$) of $0.05M$ oxalic acid is :-

(A) 100

(B) 0.2

(C) 200

(D) 400

Q.45

Match the following

	List I		List II
	(Group 13 element)		(metallic radius, pm)
A	Al	I	135
B	Ga	II	143
C	*In*	III	170
D	Tl	IV	167

The correct answer is

(A) A-I, B-II, C-IV, D-III

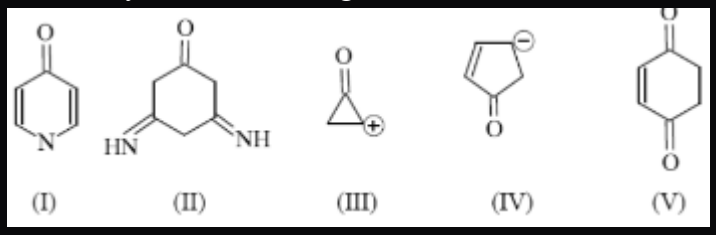
(B) A-II, B-I, C-IV, D-III

(C) A-III, B-IV, C-II, D-I

(D) A-II, B-I, C-III, D-IV

Q.46

how many of the following enolic form is more stable than keto form



Q.47

How many isomeric alcohols of molecular formula $\mathrm{C_5H_{12}O}$ give blue colour in victor - Maeyer test finally?

Q.48

At some temperature, the rate constant for the decomposition of HI on a gold surface is $0.08\mathrm{Ms}^{-1}$. The reaction is $2\mathrm{HI} \rightarrow \mathrm{H_2}(g) + \mathrm{I_2}(g)$ The time taken for the concentration of HI to drop from $1.50\mathrm{M}$ to $0.06\mathrm{M}$ is _____ (sec).

Q.49

If 5.79 A current for 20000 sec. is passed in electrolysis of 0.3 molar, 1 litre $\mathrm{CuSO_4}$ (aq) using Pt electrode then ratio of volume of gas liberated at cathode and anode at STP is $x : 1.00$. The value of x is

Q.50

When $20\ \mathrm{g}$ of naphtholic acid $(\mathrm{C_{11}H_8O_2})$ is dissolved in $50\ \mathrm{g}$ of benzene $\left(\mathrm{K_f} = 1.72\ \mathrm{K\ kg\ mol^{-1}}\right)$, a freezing point depression of $2\ \mathrm{K}$ is observed. The van't Hoff factor (i) is K, find 200K.

Q.51

If 'a' and 'b' are the first and the last terms of an A.P. as well as of an H.P. both having n terms. The product of the rth term of the first series and $(n-r+1)^{\text{th}}$ term of the second series, is

(A) Independent of both n and r and equals ab

(B) Independent on both n and r and equals a^2b

(C) independent on both n and r and equals ab^2

(D) dependent on n and r.

Q.52

Given that $\pi < \alpha < \frac{3\pi}{2}$, then the expression $\sqrt{4\sin^4\alpha + \sin^2 2\alpha} + 4\cos^2\left(\frac{\pi}{4} - \frac{\alpha}{2}\right)$ is equal to

(A) 2

(B) $2+4\sin\alpha$

(C) $2-4\sin\alpha$

(D) $2+\sin\alpha$

Q.53

For 20 observations of variable x, if $\sum(x_i-2)=20$ and $\sum(x_i-2)^2=100$, then the standard deviation of variable x is

(A) 2

(B) 3

(C) 4

(D) 9

Q.54

The value of the integral $\int_0^\infty \frac{\tan^{-1}x}{x^2+1+2x}dx$ is

(A) $\frac{\pi}{2}$

(B) $\frac{\pi}{8}$

(C) $\frac{\pi}{4}$

(D) $\frac{3\pi}{4}$

Q.55

Let R be a relation defined on the set of natural numbers N as $R=\{(x,y):x,y\in N,2x+y=41\}$. Which of the following is true?

(A) R is reflexive

(B) R is symmetric

(C) R is transitive

(D) At least one is false

Q.56

A man has to go from $(0,0)$ to $(8,8)$ by moving one step forward or upward on a 8×8 grid. If all paths are equally likely then the probability that he must pass through all the points of the form (α,α^2) & (α,α^3) is

(A) $\frac{1}{{}^{16}C_8}$

(B) $\frac{{}^2C_1\cdot{}^4C_3\cdot{}^4C_2\cdot{}^6C_2}{{}^{16}C_8}$

(C) $\frac{{}^2C_1\cdot{}^4C_3\cdot{}^4C_2}{{}^{16}C_8}$

(D) $\frac{{}^2C_1\cdot{}^4C_3}{{}^{16}C_8}$

Q.57

The value of the definite integral $\int_{-\infty}^{a} \frac{\left(\sin^{-1} e^{x}+\sec^{-1} e^{-x}\right)dx}{(\tan^{-1} e^{a}+\tan^{-1} e^{x})(e^{x}+e^{-x})}(a \in R)$ is :

(A) $\ln 3$

(B) dependent on a

(C) $\frac{\pi}{2}\ell n2$

(D) $\frac{\pi}{2}\ell n\left(2\tan^{-1} e^{a}\right)$

Q.58

Let $f(x)$ be defined by $f(x)=\begin{cases}\frac{3x-[x]}{x}, & x<1\\ \frac{3-x^{2}+x}{[x]}, & x\geq 1\end{cases}$
(where [.] is GIF) Then which is incorrect

(A) $f(x)$ is continuous at $x=1$

(B) $f(x)$ is not differentiable at $x=1$

(C) $f'\left(\frac{3}{2}\right)=-2$

(D) $f'(x)$ is continuous at $x=1$

Q.59

Solution of the differential equation $y\left(y^{3}-x\right)dx+x\left(x+y^{3}\right)dy=0$ is -

(A) $xy^{2}=x+c$

(B) $2xy=\frac{x^{2}}{y^{2}}+c$

(C) $xy^{2}=cx$

(D) $y=x+c$

Q.60

The line $\frac{x-2}{3}=\frac{y+1}{2}=\frac{z-1}{-1}$ intersects the curve $xy=c^{2}, z=0$ if c equal to

(A) ± 1

(B) $\pm\frac{1}{3}$

(C) $\pm\sqrt{5}$

(D) $\pm\sqrt{3}$

Q.61

If the system of equation

$$\begin{pmatrix} 1 & -2 & 5 \\ 2 & -1 & 1 \\ 11 & -7 & p \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \\ q \end{pmatrix}$$

has infinitely many solutions, then -

(A) $p+q=2$

(B) $p+q=10$

(C) $p-q=2$

(D) none of these

Q.62

A)	If $z=\frac{z_1+i\bar{z}_2}{z_2+i\bar{z}_1}$, then $\lvert z\rvert$ equals	p)	0
B)	If $\frac{z-2i}{z+2i}$ is purely imaginary, then $\lvert z\rvert$ equals	q)	1
C)	If $\lvert z+6\rvert=\lvert 2z+3\rvert$, then $\lvert z\rvert$ equals	r)	2
D)	Let $\omega\neq 1$ be a cube root of unity and $z=\frac{a+b\omega+c\omega^2}{c+a\omega+b\omega^2}+\frac{a+b\omega+c\omega^2}{b+c\omega+a\omega^2}$, then $\lvert z\rvert$ equals	s)	3

(A) A-r, B-p, C-s, D-r

(B) A-r, B-r, C-s, D-r

(C) $\mathrm{A-q, B-r, C-s, D-q}$

(D) A-q, B-p, C-s, D-q

Q.63

The eccentricity of the ellipse which meets the straight line $\frac{x}{7}+\frac{y}{2}=1$ on the axis of x and the straight line $\frac{x}{3}-\frac{y}{5}=1$ on the axis of y and whose axes lie along the axes of coordinates is $\frac{2\sqrt{6}}{\lambda}$. Find λ

(A) 5

(B) 7

(C) 4

(D) 9

Q.64

The sum of the roots (real or complex) of the equation $x^{2001}+\left(\frac{1}{2}-x\right)^{2001}=0$ is

(A) 2000

(B) 2001

(C) 1000

(D) 500

Q.65

The domain of the function $f(x) = {}^{16-x}C_{2x-1} + {}^{20-3x}P_{4x-5}$, where the symbols have their usual meanings, is the set

(A) $\{2, 3\}$

(B) $\{2, 3, 4\}$

(C) $\{1, 2, 3, 4\}$

(D) $\{1, 2, 3, 4, 5\}$

Q.66

Let $\mathrm{f(x)} = 5 - |\mathrm{x} - 2|$ and $\mathrm{g(x)} = |\mathrm{x} + 1|, \mathrm{x} \in \mathrm{R}$. If $\mathrm{f(x)}$ attains maximum value at α and $g(x)$ attains minimum value at β, then $\lim_{x \to -\alpha\beta} \frac{(x-1)\left(x^2-5x+6\right)}{(x^2-6x+8)}$ is equal to

(A) $\frac{1}{2}$

(B) $\frac{3}{2}$

(C) $-\frac{3}{2}$

(D) $-\frac{1}{2}$

Q.67

Two chords are drawn from the point $\mathrm{P(h, k)}$ on the circle $x^2 + y^2 = hx + ky$. If the y-axis divides both the chords in the ratio $2 : 3$, then

(A) $\mathrm{k^2 > 15\ h^2}$

(B) $\mathrm{15k^2 > h^2}$

(C) $\mathrm{h^2 = 15k^2}$

(D) None of these.

Q.68

The shortest distance between the lines $1 + x = 2y = -12z$ and $x = y + 2 = 6z - 6$ is

(A) 1 unit

(B) 4 units

(C) 2 units

(D) 3 units

Q.69

If $x = \log_3 \left[\frac{\sqrt{30+\sqrt{30+\sqrt{30+\ldots\ldots\infty}}}}{\sqrt{2+\sqrt{2+\sqrt{2+\ldots\infty}}}}\right]$, then which of the following statement hold good?

(A) $0 < x \leq 1$

(B) $1 < x \leq 2$

(C) $2 < x < 3$

(D) $x > 2$

Q.70

If $3\hat{j}, 4\hat{k}$ and $3\hat{j} + 4\hat{k}$ are the position vectors of the vertices A, B, C respectively of $\triangle ABC$, then the position vector of the point in which the bisector of $\angle \mathrm{A}$ meets BC is

(A) $\frac{5}{3}\hat{\mathrm{j}} - 4\hat{\mathrm{k}}$

(B) $5\hat{\mathrm{j}} - 4\hat{\mathrm{k}}$

(C) $5\hat{\mathrm{j}} + 4\hat{\mathrm{k}}$

(D) $\frac{5}{3}\hat{\mathrm{j}} + 4\hat{\mathrm{k}}$

Q.71

Let P be the point $(3, 2)$. Let Q be the reflection of P about the x-axis. Let R be the reflection of Q about the line $y = -x$ and let S be the reflection of R through the origin. PQRS is a convex quadrilateral. Find the area of PQRS.

Q.72

The rate at which the metal cools in moving air is proportional to the difference of temperatures between the metal and air. If the air temperature is $290\ \mathrm{K}$ and the metal temperature drops from $370\ \mathrm{K}$ to $330\ \mathrm{K}$ in 10 minutes, then the time required to drop the temperature upto $295\ \mathrm{K}$ is

Q.73

If $(\alpha\beta)(\delta\beta) = \gamma\gamma\gamma$ such that $\alpha, \beta, \delta, \gamma$ represents a number from 1 to $9 \& \alpha, \beta, \delta, \gamma$ are all different digits $\& \alpha\beta, \delta\beta$ are two digit numbers $\& \gamma\gamma\gamma$ is a three digit number, and the trace of the matrix $\mathrm{A} = \begin{bmatrix} \boldsymbol{\alpha} & 1 & 2 & 0 \\ 0 & \boldsymbol{\beta} & 1 & 1 \\ 0 & 0 & \boldsymbol{\gamma} & 3 \\ 1 & 1 & 0 & \boldsymbol{\delta} \end{bmatrix}$ is 'a', then $\left(\frac{a}{7}\right)$ is equal to

Q.74

If $y = \left(\sin^{-1} x\right)^2 + \left(\cos^{-1} x\right)^2$, then $\left(1 - x^2\right) y_2 - x y_1 =$

Q.75

In how many ways Ram can distribute 40 apples in his six children named A, B, C, D, E and F such that A gets two more than B, C gets 3 more than F and D gets five less than E and every one must have atleast one fruit

Answers & Solutions

Q.1 Answer:

—

Solution:

Given, speed of five molecules be v_1, v_2, v_3, v_4 and $v_5 = 1, 2, 3, 4$ and 5kms^{-1}

Since, root mean square speed,

$v_{\text{rms}} = \sqrt{\frac{v_1^2+v_2^2+v_3^2+v_4^2+v_5^2}{5}}$

and average speed,

$v_{\text{av}} = \frac{v_1+v_2+v_3+v_4+v_5}{5}$

On dividing Eq. (i) by Eq. (ii), we get

$$\frac{v_{\text{rms}}}{v_{\text{av}}} = \frac{\sqrt{1^2+2^2+3^2+4^2+5^2}}{1+2+3+4+5} \times \frac{5}{\sqrt{5}}$$

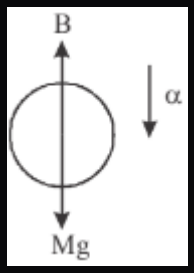
$$= \frac{\sqrt{11}}{3} \text{ or } \sqrt{11} : 3$$

Q.2 Answer:

Iron Ball

Solution:

Upthrust will be same in both the balls.



$$mg - B = ma$$

$$a = \frac{mg-B}{m} = g - \frac{B}{m}$$

Implying greater the mass greater the acceleration

$m_{\text{iron}} > m_{\text{aluminium}}$ **[for same volume]**

$\therefore a_{\text{iron}} > a_{\text{aluminium}}$

Q.3 Answer:

2×10^{20}

Solution:

$\mathrm{h}v - \phi_0 = \frac{1}{2}\mathrm{mv}_{\max}^2 . = \mathrm{KE}_{\max}$ but $\phi_0 = hv_0$ so $\mathrm{KE}_{\max}$
$= \mathrm{h}\left(v - \mathrm{v}_0\right)$
$\mathrm{v} = \frac{\mathrm{c}}{\lambda} = \frac{3 \times 10^8}{4000 \times 10^{-10}} = 7.5 \times 10^{14}\ \mathrm{Hz}$
$\therefore \mathrm{KE}_{\max} = 6.63 \times 10^{-34}\left[7.5 \times 10^{14} - 5.5 \times 10^{14}\right] = 13.26$
$\times 10^{-20}\ \mathrm{J}$

No. of photons per m^2 per sec $= \frac{\mathrm{I}}{\mathrm{h}v}$
$= \frac{100}{6.63 \times 10^{-34} \times 7.5 \times 10^{14}} = 2 \times 10^{20}$

Q.4 Answer:

$53 \pm 2\mathrm{sec}$

Solution:

$T_{\text{mean}} = \frac{53+52+55+54+51}{5} = 53\mathrm{sec}.$

$$\left.\begin{array}{l} \Delta T_1 = T_{\text{mean}} \sim T_1 = 0 \\ \Delta T_2 = 1\mathrm{sec} \\ \Delta T_3 = 2\mathrm{sec} \\ \Delta T_4 = 1\mathrm{sec} \\ \Delta T_5 = 2\mathrm{sec} \end{array}\right\} \Delta T_{\text{mean}} = \frac{0+1+2+1+2}{5} = \frac{6}{5} = 1.2\ \mathrm{sec}.$$

Q.5 Answer:

$\rho \mathrm{I}$

Solution:

The current density J, electric field E at the cross-section are related by $J = \frac{E}{\rho}$ multiplying both side by A.
$\mathrm{JA} = \frac{\mathrm{EA}}{\rho}$ **or** $\mathrm{I} = \frac{\phi}{\rho}$ **Where ϕ is electric flux.**

Q.6 Answer:

Statements I, II, III are true.

Solution:

For all metal, temperature coefficient is always positive
So (I) is true
We have

$$\mathrm{V_d} = \frac{\mathrm{i}}{\mathrm{neA}} = \frac{\frac{360\pi}{7200}}{5 \times 10^{28} \times 1.6 \times 10^{-19} \times \pi \times \left(1 \times 10^{-3}\right)^2}$$
$$= 6.25 \times 10^{-6}\ \mathrm{m/s}$$

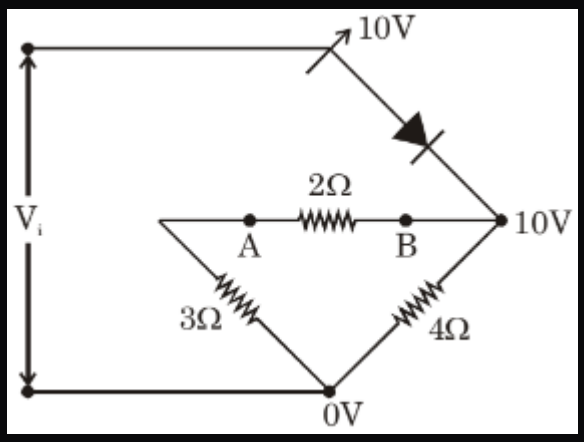
So, (II) is true
Only metals obey ohm law therefore (III) is correct.

Q.7 Answer:

4

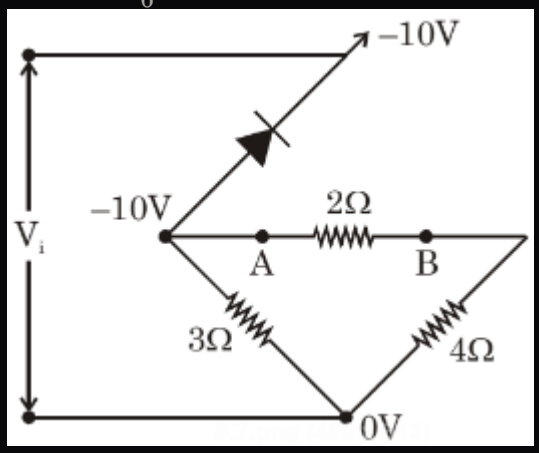
Solution:

For upper half $\mathrm{V_{AB}} = 4\ \mathrm{V}$



For lower half

$\mathrm{V_{AB}} = \frac{10}{6} \times 2 = 3.33\ \mathrm{V}$



So maximum voltage $= 4\ \mathrm{V}$

Q.8 Answer:

3

Solution:

Resistance of galvanometer,

$G = 50\Omega$

Full scale current, $i_g = 0.05$

$A = 2.97 \times 10^{-2}\ \mathrm{cm^2}$

$= 2.97 \times 10^{-2} \times 10^{-4}\ \mathrm{m^2}$

$= 2.97 \times 10^{-6}\ \mathrm{m^2}$

$i = 5\ \mathrm{A}$

$\rho = 5 \times 10^{-7}\Omega - \mathrm{m}$

Required resistance to convert the galvanometer into ammeter,

$R = \frac{i_g G}{i - i_g} = \frac{0.05 \times 50}{5 - 0.05} = \frac{25}{4.95}$

$\rho \frac{l}{A} = \frac{50}{99}$

$l = \frac{50}{99} \times \frac{A}{\rho}$

$= \frac{50}{99} \times \frac{2.97 \times 10^{-6}}{5 \times 10^{-7}}$
$= \frac{50}{99} \times \frac{29.7}{5}$
$= 10 \times 0.3$
$= 3 \text{ m}$

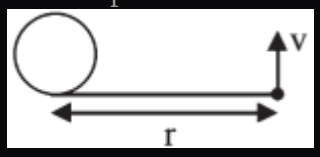
Q.9 Answer:

Tension will increase and speed will remain constant

Solution:

$\because$ Force of tension always remain $\perp$ to instantaneous velocity. So, work done by it is always zero and speed remains constant. But a given instant of time.

$\mathrm{T} = \frac{\mathrm{mv}^2}{\mathrm{r}}$



$\because$ r is decrease with time so T is increases.

Q.10 Answer:

Option (a) and (c)

Solution:

The circuit diagram is as shown below,

According to question, when given pair of point in options are connected through a resistance R, then equivalent resistance between point A and B (R_A) remains unchanged. It is only possible when current. does not flows through resistance R and circuit become Wheatstone bridge as shown in the figure below.

At the balanced condition, $\frac{P}{Q} = \frac{R}{S}$

The current flow in CD branch will be zero.

Now by checking each option, from option (a), circuit is,

Now, $\because \frac{P}{Q} = \frac{R}{S} \Rightarrow \frac{2r}{4r} = \frac{r}{2r} \Rightarrow \frac{1}{2} = \frac{1}{2}$

Hence, option is correct. From option (b), circuit is

Now, $\frac{P}{Q} = \frac{R}{S} \Rightarrow \frac{3r}{3r} \neq \frac{r}{2r}$

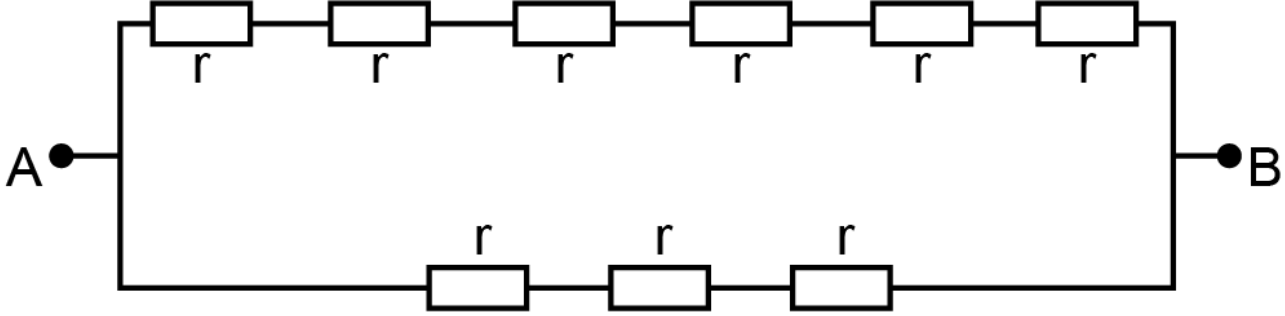
So, option (b) is also incorrect. From option (c) circuit is,

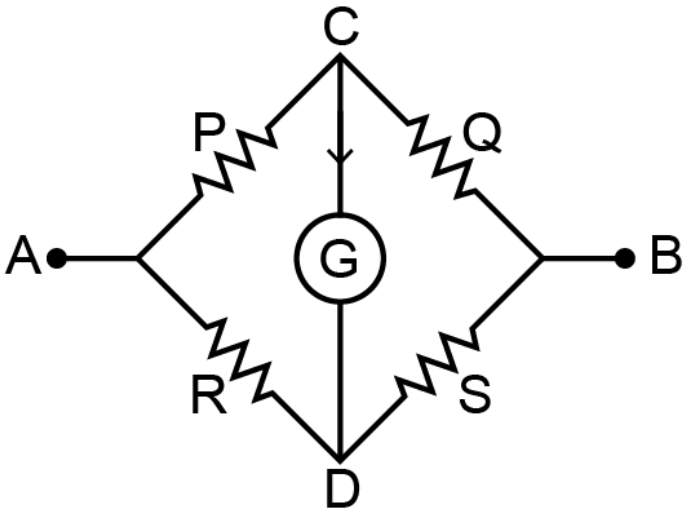
Now, $\frac{P}{Q} = \frac{R}{S} \Rightarrow \frac{4r}{2r} = \frac{2r}{r} \Rightarrow \frac{2}{1} = \frac{2}{1}$

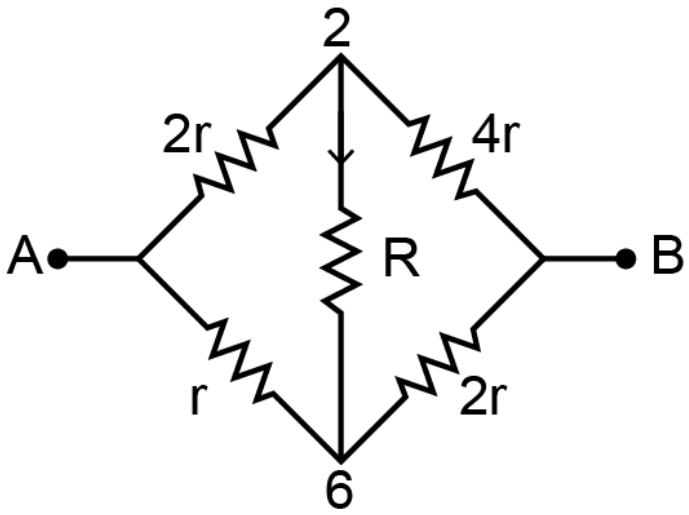
So, option (c) is also correct. From option (d), circuit is

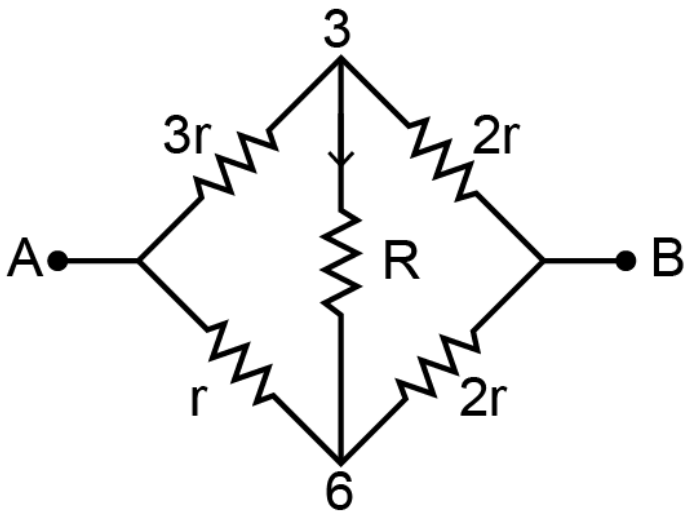
Now, $\frac{P}{Q} = \frac{R}{S} \Rightarrow \frac{4r}{2r} \neq \frac{r}{2r}$

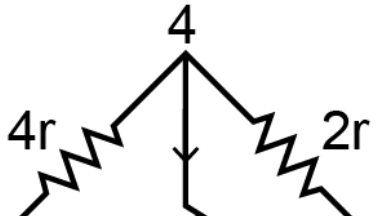
So, option (d) is also incorrect. Hence, option (a) and (c) are correct.

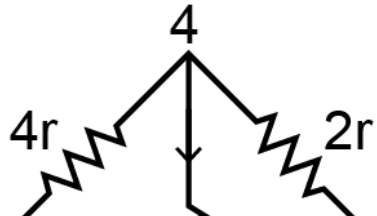

r
r
r
r
r
r
A
B
r
r
r

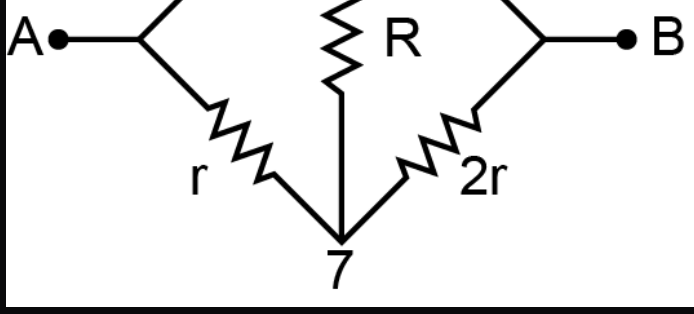

C
P
Q
A
G
B
R
S
D

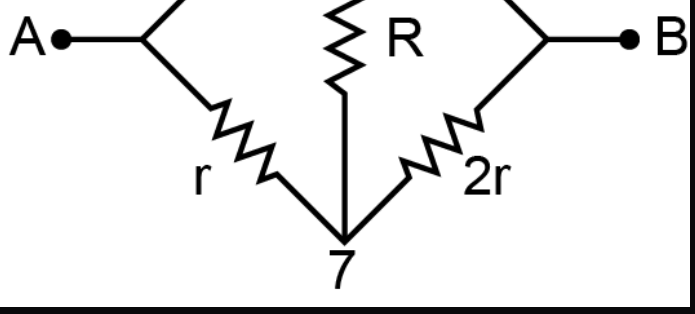

2
2r
4r
A
R
B
r
2r
6


3
3r
2r
A
R
B
r
2r
6


4
4r
2r


4
4r
2r





Q.11 Answer:

1.2

Solution:

Compare with $y = A\sin(kx - wt)$

∴ Velocity of wave $v = \frac{\omega}{k} = 2.5 \times 10^8$

∴ Refractive index

$\mu = \frac{c}{v} = \frac{3\times10^8}{2.5\times10^8} = \frac{3}{\frac{5}{2}} = \frac{6}{5}$

Q.12 Answer:

$2\ g/7$

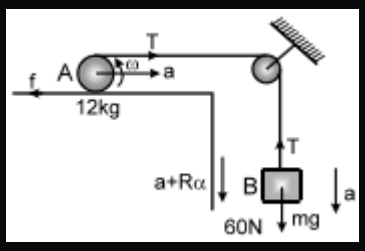
Solution:

Equation of linear motion

Block $60 - T = 6(a + R\alpha)$ **...(1)**

Cyl T-f $= 12.a$ **... (2)**

Rotation $a = R\alpha$ **...(3)**



$$\text{also } (T + f)R = \frac{MR^2}{2}\alpha$$

$$\Rightarrow \quad T + f = \frac{MR\alpha}{2} = 6R\alpha \quad \ldots(3)$$

$$60 - T = 12R\alpha \quad \ldots(4)$$

$$T - f = 12R\alpha \quad \ldots(5)$$

Multiply (5) by 2 & add all (4), (5) & (6)

$$120 = 42R\alpha = 42a$$

$$a = \frac{120}{42} = \frac{2\ g}{7} \quad [\text{Ans.: } 2\ g/7\ ]$$

Q.13 Answer:

Will decrease

Solution:

$Q = CV$

as seperation increases, capacitance decreases but V will remain same as battary is connected.

So charge will decrease.

Q.14 Answer:

4 N

Solution:

Max force applied on B, till both move together

$$F_{max} = \mu m_A g \left(1 + \frac{m_B}{m_A}\right)$$

$$= 0.5 \times 2 \times 10 \left(1 + \frac{8}{2}\right)$$

$$= 10(1 + 4) = 50 \text{ N}$$

applied force, $F = 20 \text{ N} < F_{max}$

$$\therefore \text{ both moves together, } a = \frac{F}{m_A + m_B}$$

$$= 20/(2 + 8)$$

$$= 2 \text{ m/s}^2$$

$\therefore$ **friction force on** $A = f_A = m_A a$

$= 2 \times 2 = 4 \text{ N}$

Q.15 Answer:

50 Hz

Solution:

Frequency of the current remains same, only magnitudes of current changes in a tranformer.

Q.16 Answer:

5319

Solution:

$$^{210}_{84}\text{Po} \rightarrow {}^{4}_{2}\text{He} + {}^{206}_{82}\text{Pb}$$

$$\text{mass defect } = \Delta m = (m_{po} - m_{He} - m_{Pb})$$

$$= 0.005818 u$$

$$\therefore Q = (\Delta m) \times 931.48 \text{Mev} = 5419 \text{kev}$$

From conservation of linear momentum,

$$\mathrm{Pb} = \mathrm{P}_{\alpha}$$

$$\therefore \sqrt{2\,\mathrm{m_{Pb}K_{Pb}}} = \sqrt{2\,\mathrm{m}_{\alpha}\mathrm{K}_{\alpha}} \Rightarrow \frac{\mathrm{K}_{\alpha}}{\mathrm{K_{Pb}}} = \frac{\mathrm{m_{Pb}}}{\mathrm{m}_{\alpha}} = \frac{206}{4}$$

$$\therefore \mathrm{K}_{\alpha} = (\mathrm{Q})\left(\frac{206}{206+4}\right) \quad (\mathrm{K_{total}} = \mathrm{Q})$$

$$= \frac{206}{210} \times 5419 \approx 5319\mathrm{kev}$$

Q.17 Answer:

25 m

Solution:

Given, initial velocity,

$$u = 54\ \mathrm{km/h} = 54 \times \frac{5}{18}\ \mathrm{m/s}$$

$$u = 15\ \mathrm{m/s}$$

Distance of signal from vehicle, $d = 400$ m acceleration, $a = -0.3\ \mathrm{m/s^2}$

Now, distance covered by vehicle, when it stops is given by $v^2 = u^2 + 2as$

$0 = (15)^2 - 2 \times 0.3 \times s \Rightarrow s = \frac{225}{0.6} = 375$ m

Relative position $= 400 - 375 = 25$ m

Q.18 Answer:

440 nm

Solution:

Position of minima

$$\mathrm{a}\sin\theta = \mathrm{n}\lambda$$

$$\mathrm{a}\sin\theta = \lambda_1$$

$$\sin\theta_1 = \frac{\lambda_1}{\mathrm{a}} \quad \ldots(i)$$

For maxima

$$\mathrm{a}\ \sin\theta = \frac{3\lambda_2}{2}\ [\text{approximately}] \quad \ldots(ii)$$

$$\sin\theta_1 = \frac{3}{2}\frac{\lambda_2}{\mathrm{a}}$$

$$\lambda_1 = \frac{3}{2}\lambda_2$$

$$\lambda_2 = \frac{2}{3}\lambda_1 = 440\ \mathrm{nm}$$

Q.19 Answer:

zero in all the four cases.

Solution:

As $\Delta U = nR\Delta T$ For closed path

$\Delta T = 0$

$\therefore \quad \Delta U = 0.$

Q.20 Answer:

The time taken in travelling DAB is less than that for BCD

Solution:

According to Kepler's Law of equal areas for equal times. As a planet moves in an orbit about the Sun, the areas swept out by the planet are equal for equal time intervals. So as the path DAB is close to the sun planet will move at a higher speed.

Q.21 Answer:

—

Solution:

For tangent galvanometer,

Deflection, $\theta = 30^\circ$

Resistance of coil, $R = 9\Omega$

Potential difference across coil,

$V = 4.5 \text{ V}$

Number of turns in the coil

$N = 10$

$B_H = 3.14 \times 10^{-5} \text{ T}$

Current flowing through the coil,

$I = \frac{V}{R} = \frac{4.5}{9} = 0.5 \text{ A}$

We know that, current flowing through the galvanometer is given as

$$I = \frac{2rB_H \tan\theta}{\mu_0 N}$$

$$\Rightarrow \quad r = \frac{\mu_0 NI}{2B_H \tan\theta}$$

$$= \frac{4\pi \times 10^{-7} \times 10 \times 0.5}{2 \times 3.14 \times 10^{-5} \times \tan 30^\circ}$$

$$= 10\sqrt{3} \times 10^{-2} \text{ m}$$

Q.22 Answer:

—

Solution:

Fringe width, $\beta_G = n\frac{\lambda D}{d}$

And for n number of fringes,

$\beta_G = n\frac{\lambda D}{d}$ or, $\beta_R = n\frac{\lambda D}{d}$

For $(n+1)$ th number of fringes,

$$\beta_G = (n+1)\frac{\lambda D}{d}$$

$$\because \beta_{\mathrm{R}} = \beta_{\mathrm{G}}$$

$$\text{or, } n\frac{\lambda_R D}{d} = (n+1)\frac{\lambda_R D}{d}$$

$$n.6000 \times 10^{-10}\frac{D}{d} = (n+1)5000 \times 10^{-10}\frac{D}{d} \text{ or,}$$

$$\Rightarrow 6n = (n+1)5$$

$$\Rightarrow 6n = 5n + 5 \text{ or, } 6n - 5n = 5$$

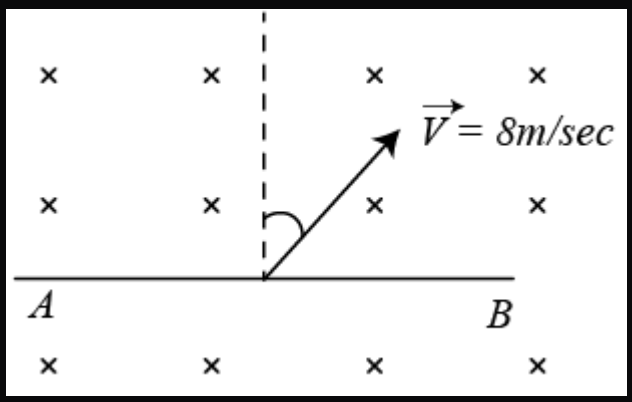
$$\therefore \quad n = 5$$

Q.23 Answer:

—

Solution:

A metal rod AB shown in the figure moves with speed $8\ \mathrm{ms}^{-1}$.



As the emf induced in the metal rod,

$$e_{AB} = V_A - V_B = l \cdot (\mathbf{V} \times \mathbf{B})$$

$$\text{But, } V_A - V_B = lvB\sin(90^\circ - 60^\circ)$$

$$V_A - V_B = 50 \times 10^{-2} \times 8 \times 2 \times \sin 30^\circ$$

$$= 8 \times \frac{1}{2} = 4\ \mathrm{V}$$

Hence, the correct option is (b).

Q.24 Answer:

—

Solution:

From Bohr model $\frac{1}{\lambda} = \mathrm{R}\left(\frac{1}{\mathrm{n}_1^2} - \frac{1}{\mathrm{n}_2^2}\right)$

$\frac{1}{\lambda_1} = \mathrm{R}\left(\frac{1}{2^2} - \frac{1}{3^2}\right) = \frac{5}{36}\mathrm{R}$

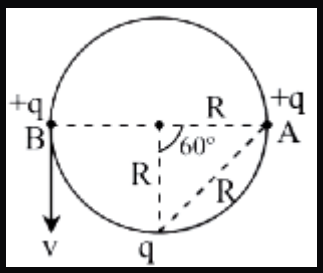
and $\frac{1}{\lambda_2} = R\left(\frac{1}{1^2} - \frac{1}{2^2}\right) = \frac{3}{4}R$

Dividing eq. (i) and (ii), we get $\frac{\lambda_1}{\lambda_2} = \frac{27}{5}$

Q.25 Answer:

—

Solution:



Energy conservation

$$\frac{1}{2}\mathrm{mv}^2+\frac{kq^2}{2R}=\frac{kq^2}{R}$$

$$\frac{1}{2}\mathrm{mv}^2=\frac{kq^2}{2R}$$

$$\mathrm{mv}^2=\frac{kq^2}{R}$$

Force on B

$$\mathrm{F}=\frac{kq^2}{4R^2}+\frac{mv^2}{R}$$

$$\mathrm{F}=\frac{kq^2}{4R}+\frac{kq^2}{R^2}\Rightarrow\frac{5kq^2}{4R^2}=\frac{5F}{4}$$

Q.26 Answer:

$\mathrm{Sc}^{3+},\mathrm{Zn}^{2+}$

Solution:

$\mathrm{Sc}^{3+}=[\mathrm{Ar}]3d^0,\mathrm{Zn}^{2+}=[\mathrm{Ar}]3s^{10}$

Both of these ions have no unpaired electron, hence these are colourless.

Q.27 Answer:

$24\mathrm{ml}$

Solution:

$$\mathrm{n_{eq}KMnO_4}=\mathrm{n_{eq}FeC_2O_4}$$

$$\text{or,}\quad\frac{\mathrm{V}\times0.2\times5}{1000}=\frac{20\times0.4\times3}{1000}$$

$$\Rightarrow\mathrm{V}=24\mathrm{ml}$$

$$\mathrm{MnO_4^-}\longrightarrow\mathrm{Mn^{2+}(n\text{-}factor}\ =5)$$

$$\mathrm{FeC_2O_4}\longrightarrow\mathrm{Fe^{2+}+CO_2(n\text{-}factor}\ =3)$$

Q.28 Answer:

$\mathrm{Mn_2O_7},\mathrm{CrO_3},\ \mathrm{V_2O_5}$

Solution:

Acidic oxides $\rightarrow Mn_2O_7, CrO_3, V_2O_5$

(due to higher oxidation state of central metal)

Basic oxide $\rightarrow Na_2O, CaO, BaO$

(due to lower oxidation state of central metal)

Alkaline and alkali metal oxide are always basic.

Neutral oxides $\rightarrow CO, NO, N_2O$

Amphoteric oxides $\rightarrow ZnO, PbO, BeO$

(acidic as well as basic)

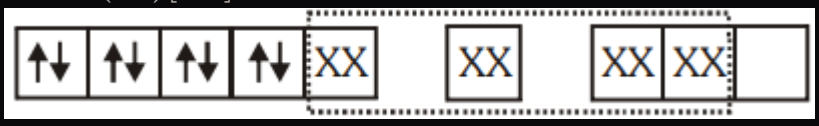
Hence, option (4) is correct.

Q.29 Answer:

$[Ni(Cl_2)(PPh_3)_2]$ has two geometrical isomers.

Solution:

(A) $Pt(78)[Xe]_{54}4f^{14}6\,s^1 5\,d^9\; Pt^{2+}6\,s^0 5\,d^8$

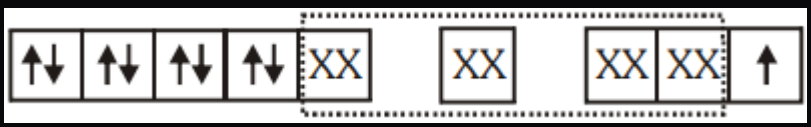


It has dsp^2 **hybridisation, square planar shape.**

$Cu(29)[Ar]4\,s^1 3\,d^{10}$

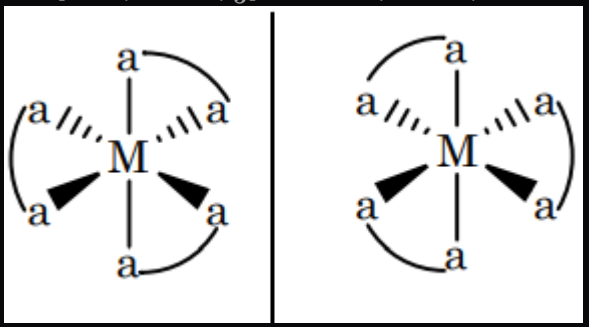
$Cu^{2+}4\,s^0 3\,d^9$

$[Cu(NH_3)_4]^{2+}$



It has dsp^2 **hybridisation, square planar shape.**

(B) $[Fe(C_2O_4)_3]^{3-}$ **is** $M(a-a)_3$ **type complex.**



The mirror images are non superimposable. Hence, optically active. (C) Synergic bonding in metal carboynyl $\uparrow, CO$ **bond strength decreases.**

(D) $[NiCl_2(PPh_3)_2$ **is tetrahedral due to steric reasons, so No geometrical isomers.**

Q.30 Answer:

Both (A) and (R) are correct and (R) is the correct explanation of (A)

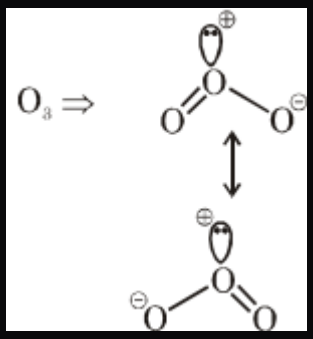
Solution:

The enthalpy of formation of a substance is
the amount of energy required to prepare it from its onstituents in their most stable states.
Since graphite is the most stable allotrope of carbon, its enthalpy of formation is zero.
Thus, both Assertion (A) and Reason (R) are correct statements and Reason (R) explains Assertion (A).

Q.31 Answer:

$O_2^+ > O_2 > O_3 > H_2O_2$

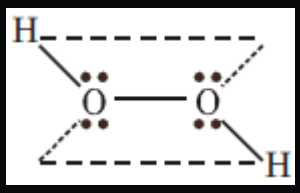
Solution:



Bond order = 1.5

$O_2^+ \Rightarrow$ **B.O.** $= \frac{N_b - N_a}{2} = \frac{10-5}{2} = 2.5$

$H_2O_2 \Rightarrow$



$\Rightarrow$ **B.O.** $= 1.0$

$O_2 \Rightarrow$ B.O. $= \frac{10-6}{2} = 2.0$

Correct order : (2) $O_2^+ > O_2 > O_3 > H_2O_2$

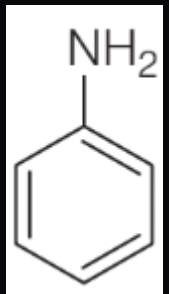
Q.32 Answer:

II, III, IV

Solution:

Key Idea The groups in which nitrogen contain double bond or present in heterocyclic ring can not be estimated by Kjeldahl's method.

Structure of aniline is

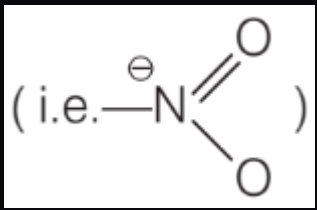


as nitrogen in

aniline does not contain double bond, it can be estimated by Kjeldahl's method.

(II) Azobenzene contain $-\mathrm{N}=\mathrm{N}-$ group, thus does not estimated by Kjeldahl's method.

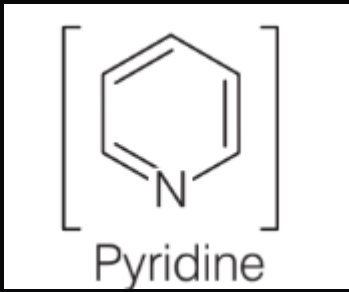
(III) In nitrobenzene, nitrogen also contains



double bond, thus can not be

estimated by Kjeldahl's method.

(IV) Pyridine is a heterocyclic organic compound in which nitrogen is present in cyclic ring.



Thus, it is also not estimated by Kjeldahl's method. Hence, option (a) is the correct answer.

Q.33 Answer:

$\frac{a_0}{2Z}, \frac{3a_0}{Z}$

Solution:

Probability of finding the electron is

$$\psi^2 = 0 \quad \text{or } \psi = 0 \quad \sigma = \frac{2Zr}{a_0}$$

$$(\sigma - 1) = 0 \Rightarrow \sigma = 1 \quad r = \frac{\sigma a_0}{2Z}$$

$$\text{or } r = \frac{a_0}{2Z} \quad \text{or } \sigma^2 - 8\sigma + 12 = 0$$

$$\sigma = 6; \quad \sigma = 2$$

$$\text{if } \sigma = 6 \quad \Rightarrow r = \frac{6a_0}{2Z} = \frac{3a_0}{Z}$$

$$\sigma = 2 \quad r = \frac{2a_0}{2Z} = \frac{a_0}{Z}$$

$$\sigma = 1 \quad r = \frac{a_0}{2Z} \quad = \frac{a_0}{2Z}$$

Q.34 Answer:

1,2

Solution:

1. $CH_3CH_2CH=NOH \xrightarrow[-2CH_3COOH]{(CH_3CO)_2O} \underset{\text{(Ethyl cyanide)}}{CH_3CH_2CN}$

2. $CH_3CH_2Cl \xrightarrow[-\ KCl]{\text{Enthanolic KCN}} \underset{\text{(Major)}}{CH_3CH_2CN}$

3. $CH_3CH_2Cl \xrightarrow[-AgCl]{\text{Ethanolic AgCN}} \underset{\text{(Major)}}{CH_3CH_2CN}$

4. $CH_3CH_2-\overset{\overset{O}{\|}}{C}-N\begin{matrix}CH_3\\CH_3\end{matrix} \xrightarrow{C_6H_5SO_2Cl,\ Py\ :\ 70^\circ C} \underset{\text{No reaction}}{}$

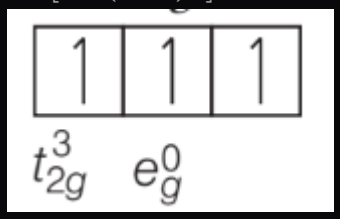
So, reactions 1 and 2 give ethyl cyanide as the major product.

Q.35 Answer:

$[Co(CN)_6]^{3-}$

Solution:

(I) $[Cr(CN)_6]^{3-}$, $Cr^{3+} \Rightarrow 3d^3$ $\therefore$ CN **is a strong field ligand.**

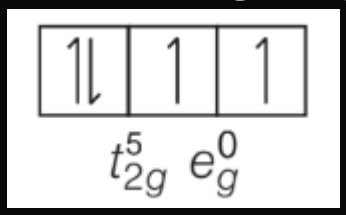


$\therefore$ **Number of unpaired** $e^- = 3$

$\Rightarrow N = \sqrt{n(n+2)} = \sqrt{3(3+2)} = \sqrt{15}BM$

(II) $[Mn(CN)_6]^{3-}$; $Mn^{3+} \Rightarrow 3d^4$

$\because$ CN **is a strong field ligand.**

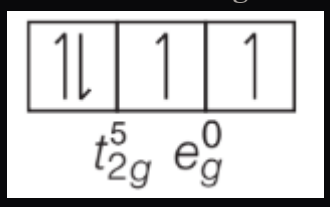


$\therefore$ **Number of unpaired** $e^- = 2$

$\Rightarrow \quad N = \sqrt{2(2+2)} = \sqrt{8}BM$

(III) $[Fe(CN)_6]^{3-}$; $Fe^{3+} \rightarrow 3d^5$

$\because$ CN **is a strong field ligand.**

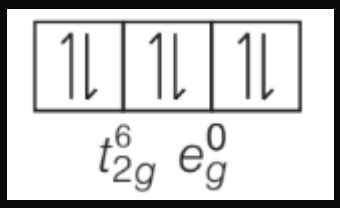


$\therefore$ **Number of unpaired** $e^- = 2$

$N = \sqrt{2(2+2)} = \sqrt{8}BM$

(IV) $[Co(CN)_6]^{3-}$, $Co^{3+} \longrightarrow 3d^6$

$\because CN^-$ is a strong field ligand.



$\therefore$ Number of unpaired $e^- = 0$

$\Rightarrow \quad N = 0$

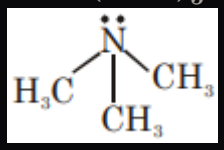
Hence, minimum value of magnetic moment is for $[Co(CN)_6]^{3-}$.

Q.36 Answer:

$CH_3\ N\ (SiH_3)_2$

Solution:

(1) $\ddot{N}\ (CH_3)_3 \Rightarrow NX_3$ L type

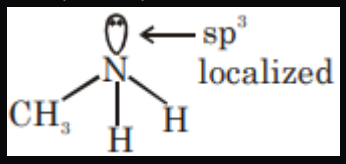


lone pair sp^3 hybrid orbital

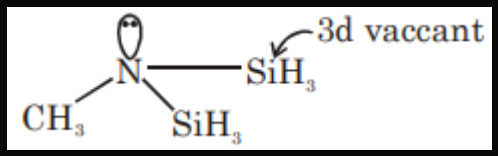
$\therefore$ localised

(2) $[N\ (CH_3)_4]^{\oplus} \Rightarrow NX_4^+$ type (X)

(3) $(CH_3)\dot{N}_2 \Rightarrow NX_3$ L type



(4)



$\therefore$ due to $p_\pi - d_\pi$ back bonding lone pair is delocalized in the orbitals of it.

Q.37 Answer:

(i) and (ii)

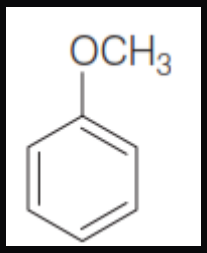
Solution:

(i) The Cl group activates benzene ring more by resonance effect.

Cl group being more electronegative withdraws electrons from benzene ring and shows - I effect.

(ii) The methyl group of ether can donate electrons through inductive effect that is not as strong as electron donation effect to the pi-system.

So, resonance effect dominates over inductive effect in anisole.



(iii) p-nitrobenzoic acid is more acidic than m-nitrobenzoic acid. When both acids donate H^+ from $-COOH$ group, then p-nitrobenzoate is more stabilised due to m-effect of nitro group at meta position by deactivating the ring.

So, statement (iii) is incorrect.

(iv) Diphenylamine is less basic than aniline. The lone pair of N-atom in diphenylamine is delocalised on two benzene rings due to which it is less available to donate resulting into decrease in basicity. In aniline, it is delocalised on one benzene ring, so lone pair is more available for donation. Hence, it is more basic.

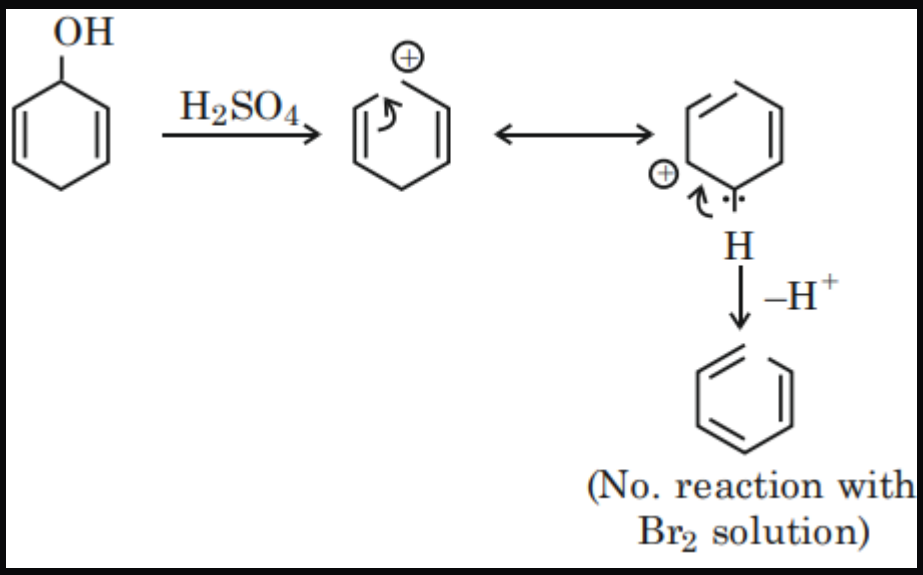
So, statement IV is incorrect.

Q.38 Answer:

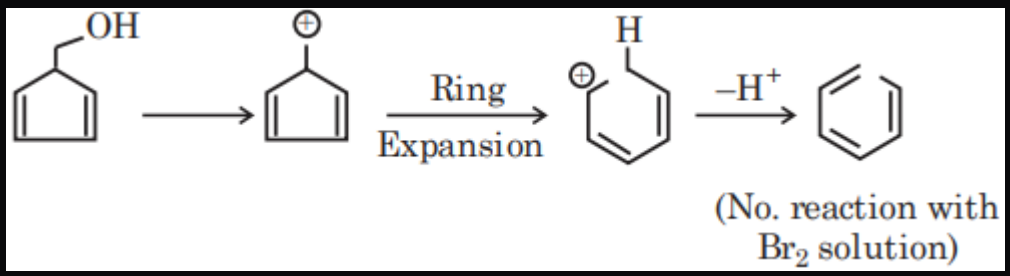
$Me_3C - CH_2 - CH_2OH$

Solution:

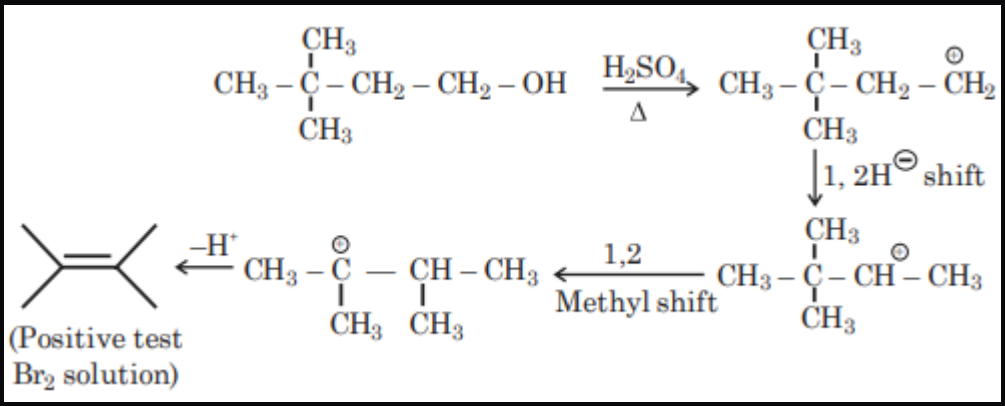
(A)



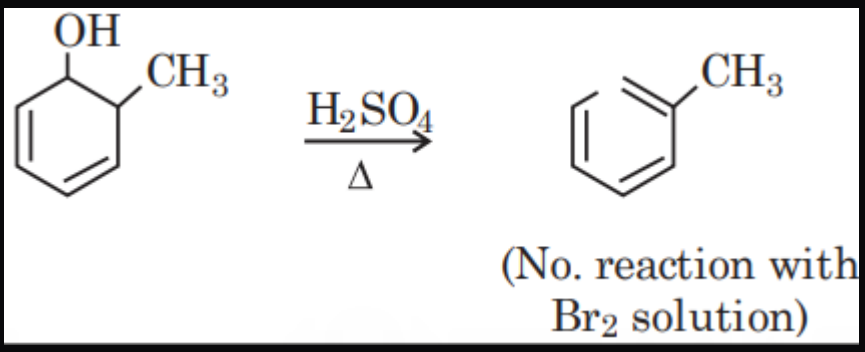
(B)



(C)



(D)



Q.39 Answer:

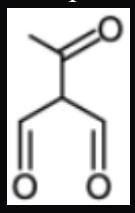
Variable oxidation states

Solution:

Catalytic activity of transition elements is due to their variable oxidation states and to form complexes

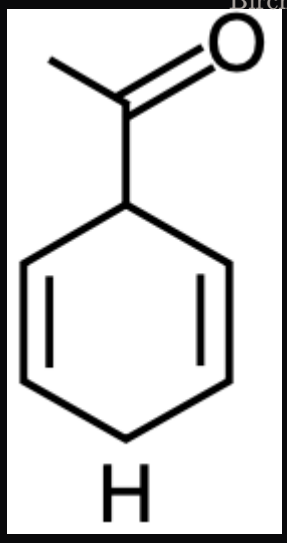
Q.40 Answer:

Compound J is

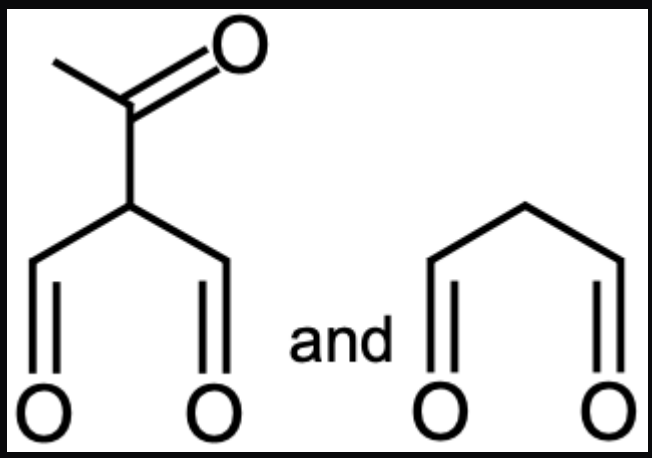


Solution:

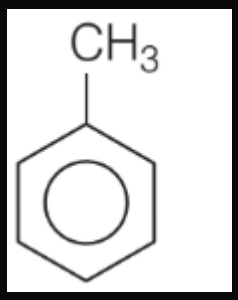
Acetophenone $\xrightarrow[\text{Birch reduction}]{\text{Na, Liq. } NH_3}$



(obtained by 1,4 reduction at ipso-para position to keto group because keto group is electron withdrawing group) H forms ozonide intermediate (I) with ozone, which undergoes reduction by Zn to form

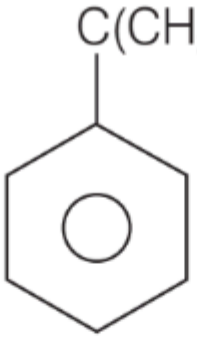


Q.41 Answer:

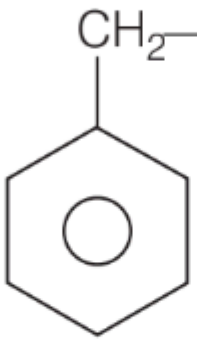


Solution:

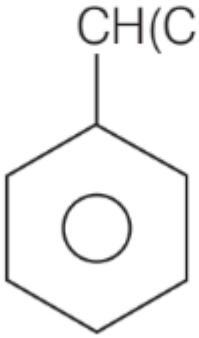
Key Idea More be the H-atoms associated (bonded) with the C-atom, which is bonded with the benzene ring, more will be the number of 'no bond resonance' structures shown by them.

In structure (a) 

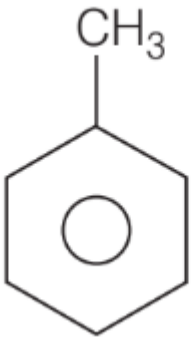
, the C-atom associated with the ring has no H-atom.

In structure (b) 

, the C-atom bonded with the ring has 2 H-atoms.

In structure (c) 

, the C-atom bonded with the ring has one H-atom.

In structure (d) 

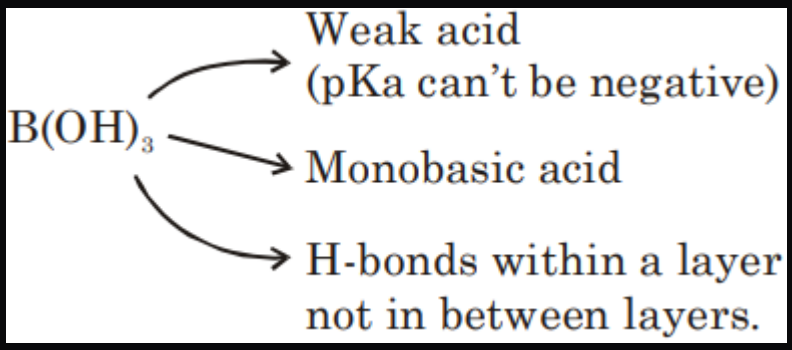
, the C-atom bonded with the ring has 3H-atoms.

Hence, option (d) is the correct answer.

Q.42 Answer:

Cis-diols can be used to increase acidic strength in its aq. solution.

Solution:



Q.43 Answer:

I, II, III & IV

Solution:

$$Zn(OH)_2 \rightleftharpoons Zn_s^{+2} + 2OH_{2\,S}$$

$$S = (k_{sp}/4)^{1/3} = 10^{-4}$$

In presence of $0.01 M Zn^{+2}$.

$$Zn(OH)_2 \rightleftharpoons \underset{S'+0.01}{+2} + \underset{2\,S'}{2OH^-}$$

$$k_{sp} = 0.01 \times (2\,S)^2$$

$$S = \sqrt{4 \times 10^{-12}/10^{-2} \times 4} = 10^{-5}$$

Q.44 Answer:

100

Solution:

Specific conductance,

$$K = \frac{1}{R} \cdot \frac{\ell}{A} = 10^{-2}\ S\ cm^{-1}$$

∴ Equivalent conductance, $\wedge_{eq} = \frac{K \times 1000}{N}$

$$= \frac{10^{-2}}{10^{-1}} \times 10^3$$

$$= 10^2\ S\ cm^2 eq^{-1}$$

Q.45 Answer:

A-II, B-I, C-IV, D-III

Solution:

The atomic radii or metallic radii of elements increases down the group due to increasing shells with the exception of Gallium (Ga) **in the** 13th **group.**

This is due to the presence of d-subshell that offers poorer shielding of the outermost p-electrons.

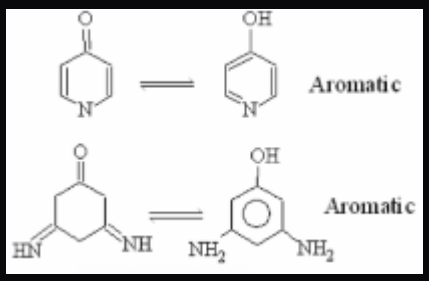
Thus, the order of metallic radius will be:-

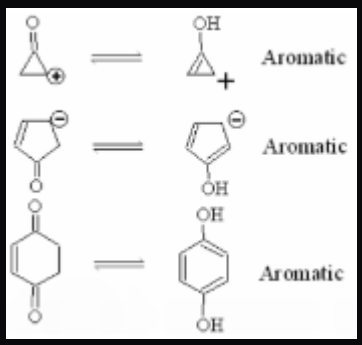
$$Ga < Al < In < Tl$$

Q.46 Answer:

—

Solution:

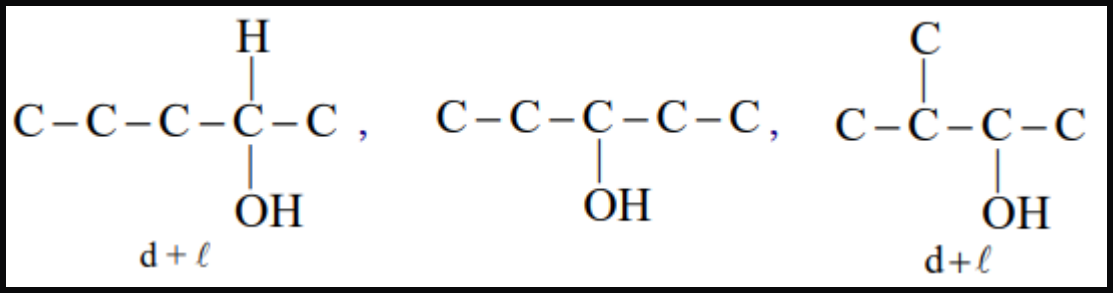




Q.47 Answer:

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Solution:



Q.48 Answer:

—

Solution:

Reaction follows zero order kinetics (Note the unit of k) $2A \rightarrow$ products

$$k_0 = \frac{x}{nt}$$

$$k_0 = \frac{a-(a-x)}{nt}$$

$$0.08 = \frac{1.50-0.06}{2t}$$

$$t = 9\text{sec}$$

Q.49 Answer:

—

Solution:

$\mathrm{n_e} = \frac{5.79 \times 20000}{96500} = 1.2 \text{ mole}$

Cathode :

$\mathrm{Cu^{+2}} + 2\mathrm{e} \longrightarrow \mathrm{Cu}$

$0.3 \text{ mole} \Rightarrow 0.6 \text{ mole}$

$2\mathrm{H_2O}(l) + 2\mathrm{e} \longrightarrow \mathrm{H_2} \uparrow + 2\mathrm{OH}$

$0.6 \text{ mole} \Rightarrow \quad 0.3 \text{ mole}$

Anode:

$2\mathrm{H_2O}(l) \rightarrow \underset{\frac{1.2}{4} \text{ mole}}{\mathrm{O_2}} + 4\mathrm{H^+} + \underset{\Leftarrow 1.2 \text{ mole}}{4\mathrm{e^-}}$

$\mathrm{V_{O_2}} = \frac{1.2}{4} = 0.3 \text{ mole}$

$\frac{\mathrm{V_{H_2}}}{\mathrm{V_{O_2}}} = \frac{0.3}{0.3} = 1$

Q.50 Answer:

—

Solution:

$\Delta \mathrm{T_f} = \mathrm{i} \times \mathrm{kf} \times \mathrm{m}$

$2 = \mathrm{i} \times 1.72 \times \frac{\frac{20}{\mathrm{M_{solute}}}}{\frac{50}{1000}}$

$2 = \mathrm{i} \times 1.72 \times \frac{20}{172} \times \frac{1000}{50}$

$\mathrm{i} = \frac{1}{2}$

Q.51 Answer:

Independent of both n and r and equals ab

Solution:

$$a, (a+d), a+2d), \ldots\ldots\ldots\ldots\ldots \underbrace{\text{b}}_{\text{n}^{\text{th}}\text{ term}}$$

$$\therefore \quad b = a + (n-1)d \Rightarrow d = \frac{b-a}{n-1}$$

$$T_r \text{ or A.P. } = a + (r-1)d$$

$$= a + \frac{(r-1)(b-a)}{n-1} = \frac{an - a + (r-1)(b-a)}{(n-1)} = \frac{an + r(b-a) - b}{n-1}$$

where $\frac{1}{A}, \frac{1}{A+D}, \ldots\ldots\ldots\ldots\ldots\ldots\ldots \frac{1}{A+(n-1)D}$

$$\therefore \quad A + (n-1)D = \frac{1}{b};$$

$$\therefore \quad \frac{1}{a} + (n-1)D = \frac{1}{b};$$

Q.52 Answer:

2

Solution:

$$\sqrt{4\sin^4\alpha + 4\sin^2\cos^2\alpha} + 2 \cdot 2\cos^2\left(\frac{\pi}{4} - \frac{\alpha}{2}\right)$$

$$= \sqrt{4\sin^2\alpha\left(\sin^2\alpha + \cos^2\alpha\right)} + 2 \cdot \left(1 + \cos\left(\frac{\pi}{2} - \alpha\right)\right)$$

$$= |2\sin\alpha| + 2 + 2\sin\alpha$$

$$= -2\sin\alpha + 2 + 2\sin\alpha = 2$$

Q.53 Answer:

2

Solution:

Note that standard derivation is independent of change of origin.

$$\therefore \quad \text{S.D. of } x_\text{i} = \text{S.D. of } (x_\text{i} - 2)$$

$$\therefore \text{ S.D. of } (x_i - 2) = \sqrt{\frac{1}{n}\sum_{\text{i}}^{20}(x_i - 2)^2 - \left[\frac{\sum(x_i - 2)}{n}\right]^2}$$

$$= \sqrt{\frac{100}{20} - (1)^2}$$

$$\Rightarrow ARKSMS = 2$$

$$\Rightarrow \text{ Required } S.D = 2$$

Q.54 Answer:

$\frac{\pi}{4}$

Solution:

$I = \int_0^\infty \frac{\tan^{-1} x}{(x+1)^2} dx$

Put $x = \frac{1}{t} \Rightarrow dx = -\frac{1}{t^2} dt$

$$= \int_\infty^0 \frac{\tan^{-1}\left(\frac{1}{t}\right)}{\left(\frac{1}{t}+1\right)^2} \left(\frac{-1}{t^2}\right) dt = \int_0^\infty \frac{\cot^{-1} t}{(1+t)^2} dt = I$$

$$2I = \frac{\pi}{2} \int_0^\infty \frac{1}{(1+t)^2} dt \Rightarrow I = \frac{\pi}{4}$$

Q.55 Answer:

At least one is false

Solution:

We have $R = \{x, y) : 2x + y = 41, x, y, \in N\}$

$2x + y = 41 \quad \Rightarrow \quad x = \frac{41-y}{2}$

$y = 1 \quad \Rightarrow \quad x = 20 \in N \quad \therefore \quad (20, 1) \in R$

$y = 2 \quad \Rightarrow \quad x = 39/2 \notin R \quad \therefore \quad (39/2, 2) \notin R$

Similarly, (19, 3), (18, 5), (17, 7), (16, 9), (15, 11), (14. 13), (13, 15), (12, 17), (11, 19), (10, 21), (9, 23), $(8, 25), (7, 27), (6, 29), (5, 31), (4, 33), (3, 35), (2, 37), (1, 39)$ are in R

R is not reflexive, because $1 \in N$ and $(1, 1) \notin R$.

R is not symmetric, because $(20, 1) \in R$ but $(1, 20) \notin R$.

R is not transitive, because $(19, 3)(3, 35) \in R$ but $(19, 35) \notin R$.

Q.56 Answer:

$\frac{^2C_1 \cdot {}^4C_3}{^{16}C_8}$

Solution:

In grid of 8×8 points, $(1, 1), (2, 4)$ and $(2, 8)$ are of types (α, α^2) and (α, α^3)

$\Rightarrow$ from $(0, 0)$ to $(8, 8)$ via $(1, 1), (2, 4)$ and $(2, 8)$

$(0,0) \xrightarrow{2!} (1,1) \xrightarrow{\frac{4!}{3!}} (2,4) \xrightarrow{1} (2,8) \xrightarrow{1} (8,8)$

$\Rightarrow$ Number of favourable ways

$= 2! \times \frac{4!}{3!} \times 1 \times 1 = {}^2C_1 \cdot {}^4C_1$

Total number of possible ways from $(0, 0)$ to

$(8, 8) = \frac{16!}{8!8!} = {}^{16}C_8$

Q.57 Answer:

$\frac{\pi}{2} \ell n 2$

Solution:

We have

$$I = \int_{-\infty}^{a} \left(\frac{\sin^{-1} e^x + \cos^{-1} e^x}{\tan^{-1} e^a + \tan^{-1} e^x}\right) \left(\frac{e^x}{e^{2x}+1}\right) dx$$

$$= \frac{\pi}{2} \int_{-\infty}^{a} \frac{1}{\left(\tan^{-1} e^a + \tan^{-1} e^x\right)} \left(\frac{e^x}{(e^{2x}+1)}\right) dx$$

Put $\tan^{-1} e^x = t \Rightarrow \frac{e^x}{e^{2x}+1}dx = dt$

$$I = \frac{\pi}{2}\int_0^{\tan^{-1} e^a} \frac{dt}{(t + \tan^{-1} e^a)}$$

$$= \frac{\pi}{2}\left[\ell n\left(t + \tan^{-1} e^a\right)\right]_0^{\tan^{-1} e^a}$$

$$= \frac{\pi}{2}\left[\ell n\left(2\tan^{-1} e^a\right) - \ell n\left(\tan^{-1} e^a\right)\right]$$

$$= \frac{\pi}{2}\ln 2 \text{ Ans.}\ ]$$

Q.58 Answer:

$f'(x)$ is continuous at $x = 1$

Solution:

$$f(x) = \begin{cases} 3 & x < 1 \\ 3 & x = 1 \\ 3 - x^2 + x & x < 1 \end{cases}$$

$$\text{L.L} = \text{R.L} = f(1) \Rightarrow \text{Cont. at } x = 1$$

$$f'(x) = \begin{cases} 0, & x < 1 \\ -2x + 1 & x > 1 \end{cases}$$

$$\text{L.D} = 0.\ \text{R.D} = -1$$

$\therefore$ **Not differentiable at** $x = 1$

$\therefore f^1(x)$ **is discontinuous at** $x = 1$

Also $f^1\left(\frac{3}{2}\right) + f^1(0) = -2\left(\frac{3}{2}\right) + 1 + 0 = -2$

Q.59 Answer:

$2xy = \frac{x^2}{y^2} + c$

Solution:

$$y^3(ydx + xdy) = xydx - x^2dy$$

$$y^3 d(xy) = x\frac{(ydx - xdy)}{y^2}$$

$$\int yd(xy) = \int \frac{x}{y} d\left(\frac{x}{y}\right)$$

$$xy = \frac{1}{2}\left(\frac{x}{y}\right)^2 + c$$

$$2xy = \frac{x^2}{y^2} + c$$

Q.60 Answer:

$\pm\sqrt{5}$

Solution:

We have $z = 0$ for the point where the line intersecte the curve

$$\frac{x-2}{3} = \frac{y+1}{2} = \frac{0-1}{-1}$$

$$x = 5, y = 1$$

$$c^2 = 5$$

$$c = \pm\sqrt{5}$$

Q.61 Answer:

$p - q = 2$

Solution:

$$x - 2y + 5z = 3$$

$$2\mathrm{x} - \mathrm{y} + \mathrm{z} = 1$$

$$11x - 7y + pz = q$$

$$\Delta = 0 \Rightarrow \begin{vmatrix} 1 & -2 & 5 \\ 2 & -1 & 1 \\ 11 & -7 & \mathrm{p} \end{vmatrix} = 0$$

$$(-p + 7) + 2(2p - 11) + 5(-14 + 11) = 0$$

$$\Rightarrow \quad \mathrm{p} = 10$$

$$\Delta_{\mathrm{x}} = 0 \Rightarrow \begin{vmatrix} 3 & 2 & 5 \\ 1 & 1 & 1 \\ \mathrm{q} & 7 & 10 \end{vmatrix} = 0$$

$$\Rightarrow 3(10 - 7) - 2(10 - q) + 5(7 - q) = 0$$

$$\Rightarrow \mathrm{q} = 8$$

$$p - q = 2$$

Q.62 Answer:

$\mathrm{A} - \mathrm{q}, \mathrm{B} - \mathrm{r}, \mathrm{C} - \mathrm{s}, \mathrm{D} - \mathrm{q}$

Solution:

(A) $|z| = \frac{|z_1 + i\bar{z}_2|}{|z_2 + i\bar{z}_1|} = \frac{|i||-iz_1 + \bar{z}_2|}{|z_2 + iz_1|} = 1$

(B) Here $\mathrm{Arg}\left(\frac{z-2i}{z+2i}\right) = \frac{\pi}{2}(\text{ or }) - \frac{\pi}{2}$,

So it represents a semicircle with diametric ends at points $(0, 2)$ & $(0, -2)$

So, $|z|$ **= radius** $= 2$

(C) $(x + 6)^2 + y^2 = (2x + 3)^2 + (2y)^2 \Rightarrow x^2 + y^2 + 12x + 36 = 4x^2 + 4y^2 +$

$12\mathrm{x} + 9$

$\Rightarrow 3\mathrm{x}^2 + 3\mathrm{y}^2 = 27 \Rightarrow \mathrm{x}^2 + \mathrm{y}^2 = 9, |\mathrm{z}| = 3$

(D) $z = \frac{1}{\omega}\frac{a\omega + b\omega^2 + c\omega^3}{c + a\omega + b\omega^2} + \frac{1}{\omega^2}\frac{a\omega^2 + b\omega^3 + c\omega^4}{b + c\omega + a\omega^2} = \frac{1}{\omega} + \frac{1}{\omega^2} = -1$

Q.63 Answer:

7

Solution:

$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$
$\frac{x}{7}+\frac{y}{2}=1$ **meets** x**-axis at** $A(7,0)$**, line** $\frac{x}{3}-\frac{y}{5}=1$ **meets** y**-axis at** $B(0,-5)$.
(1) passes through A & B

$$\Rightarrow \frac{49}{a^2}+0=1, 0+\frac{25}{b^2}=1$$

$$\Rightarrow a^2=49, b^2=25; b^2=a^2\left(1-e^2\right) \Rightarrow 25=49\left(1-e^2\right)$$

$$\Rightarrow \mathrm{e}^2=\frac{24}{49} \Rightarrow \mathrm{e}=\frac{2\sqrt{6}}{7}$$

Q.64 Answer:

500

Solution:

$$x^{2001}-\left[{}^{2001}C_0 x^{2001}-{}^{2001}C_1 x^{2000}\cdot\frac{1}{2}+{}^{2001}C_2 x^{1999}\cdot\frac{1}{4}\ldots\ldots\ldots\right]=0$$

$${}^{2001}\mathrm{C}_1\mathrm{x}^{2000}\cdot\frac{1}{2}-{}^{2001}\mathrm{C}_2\mathrm{x}^{1999}\cdot\frac{1}{4}\ldots\ldots\ldots\ldots=0$$

$$\text{sum of the roots } ={}^{2001}\mathrm{C}_2\cdot\frac{1}{4}\cdot\frac{2}{{}^{2001}\mathrm{C}_1}=\frac{2001\cdot 2000}{4\cdot 2001}=500$$

Q.65 Answer:

$\{2,3\}$

Solution:

Ans. (1) For $f(x)$ **to be defined**
(i) $16-x>0 \Rightarrow x<16$
(ii) $2x-1\geq 0 \Rightarrow x\geq \frac{1}{2}$
(iii) $20-3x>0 \Rightarrow x<\frac{20}{3}$
(iv) $4x-5\geq 0 \Rightarrow x\geq \frac{5}{4}$
(v) $16-x\geq 2x-1 \Rightarrow x\leq \frac{17}{3}$
(vi) $20-3x\geq 4x-5 \Rightarrow x\leq \frac{25}{7}$
(vii) $16-x$ **is an integer,**
$\therefore$ x **should be an integer**
$\therefore \frac{5}{4}\leq x\leq \frac{25}{7}$ **and** $x\in I \Rightarrow x=2,3$
$\therefore$ **Domain of** $f=\{2,3\}$.

Q.66 Answer:

$\frac{1}{2}$

Solution:

$\mathrm{f(x)}=5-|\mathrm{x}-2|$ **is maximum at** $\mathrm{x}=2 \Rightarrow \alpha=2$
$\mathrm{g(x)}=|\mathrm{x}+1|$ **is minimum at** $\mathrm{x}=-1 \Rightarrow \beta=-1$

Now $\lim_{x\to-\alpha\beta}\frac{(x-1)\left(x^2-5x+6\right)}{(x^2-6x+8)}=\lim_{x\to-(2)(-1)}\frac{(x-1)(x-2)(x-3)}{(x-2)(x-4)}$

$$=\lim_{x\to2}\frac{(x-1)(x-3)}{x-4}$$

$$=\frac{(2-1)(2-3)}{2-4}$$

$$=\frac{1}{2}$$

Q.67 Answer:

$\mathrm{k}^2>15\,\mathrm{h}^2$

Solution:

Point $\mathrm{P}(\mathrm{h},\mathrm{k})$ is lying on the circle. Let the y axis divides the chords in the ratio $2:3$ at the point $(0,\beta)$. Then co-ordinates of the other end of the chord will be $\left(-\frac{3h}{2},\frac{5\beta-3k}{2}\right)$. This point will lie on the circle.

$\Rightarrow$ for two distinct values of $\beta, 64\mathrm{k}^2-60\left(\mathrm{h}^2+\mathrm{k}^2\right)>0\Rightarrow4\mathrm{k}^2>60\,\mathrm{h}^2\Rightarrow\mathrm{k}^2>15\,\mathrm{h}^2$

Q.68 Answer:

2 units

Solution:

$\frac{x+1}{1}=\frac{y}{1/2}=\frac{z}{-1/12};\frac{x}{1}=\frac{y+2}{1}=\frac{z-1}{1/6}$

$(1,1/2,-1/12);(1,1,1/6)\quad\longrightarrow(-1,0,0);(0,-2,1)$

$(12,6,-1);(6,6,1)$

$$\vec{n}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\12&6&-1\\6&6&1\end{vmatrix}=12\hat{i}-18\hat{j}+36\hat{k}=6(2\hat{i}-3\hat{j}+6\hat{k})$$

$$d=\left|\frac{(\vec{a}_1-\vec{a}_2)\cdot\vec{n}}{|\vec{n}|}\right|=\left|\frac{(\hat{i}-2\hat{j}+k)\cdot(2\hat{i}-3\hat{j}+6\hat{k})}{\sqrt{2^2+3^2+6^2}}\right|$$

$$d=\frac{2+6+6}{7}=\frac{14}{7}=2$$

Q.69 Answer:

$0<x\leq1$

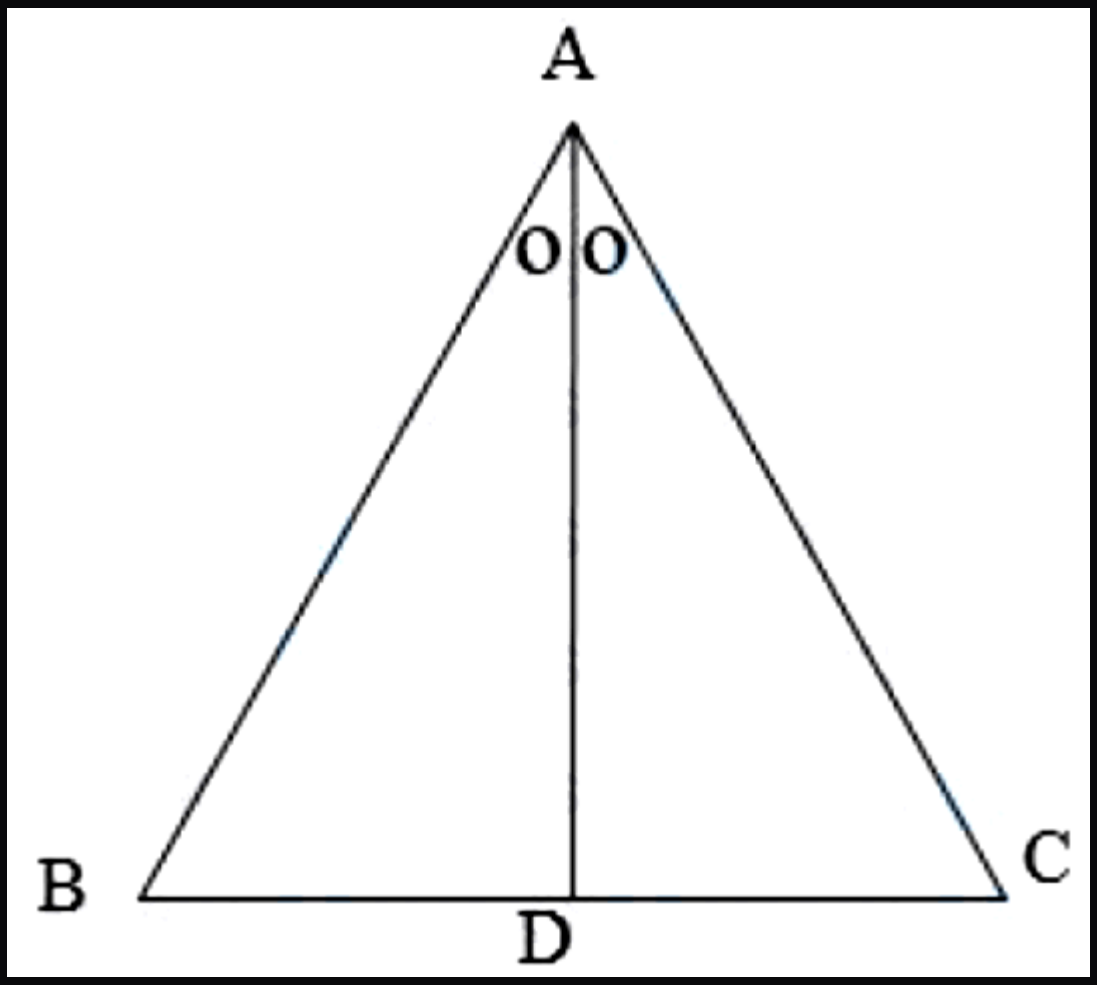
Solution:

$$t_1 = \sqrt{30 + \sqrt{30 + \sqrt{30}}} \Rightarrow t_1 = \sqrt{30 + t_1}$$
$$\Rightarrow t_1^2 - t_1 - 30 = 0$$
$$t_1 = 6$$
$$t_2 = \sqrt{2 + \sqrt{2 + \ldots\ldots}}$$
$$\Rightarrow t_2 = \sqrt{2 + t_2} = t_2^2 = 2 + t_2$$
$$t_2^2 - t_2 - 2 = 0 \Rightarrow t_2 = 2$$
$$x = \log_3 \left(\frac{6}{2}\right) = 1$$

Q.70 Answer:

$\frac{5}{3}\hat{j} + 4\hat{k}$

Solution:



We have $\overline{a} = 3\hat{j}, b = 4\hat{k}$ **and** $\overline{c} = 3\hat{j} + 4\hat{k}$

Angle bisector AD **divides** BC **in the ratio** $AB : AC$

$AB = \sqrt{9 + 16} = 5$ and $AC = \sqrt{0 + 16} = 4$

Thus D divides BC in the ratio $5:4$

$\therefore \mathrm{D}=\left[0, \frac{5(3)}{5+4}, \frac{5(4)+4(4)}{5+4}\right]$

$=\left(0, \frac{15}{9}, \frac{20+16}{9}\right)=\left(0, \frac{5}{3}, 4\right)=\frac{5}{3}\hat{\mathrm{j}}+4\hat{\mathrm{k}}$

Q.71 Answer:

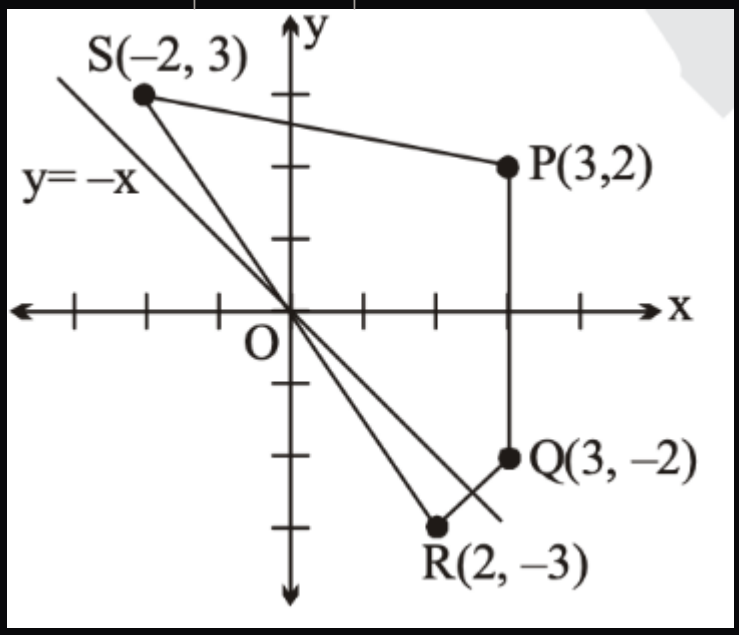
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Solution:

We have $\overrightarrow{\mathrm{d}}_1=\overrightarrow{\mathrm{RP}}=\hat{\mathrm{i}}+5\hat{\mathrm{j}}$

$\overrightarrow{\mathrm{d}}_2=\overrightarrow{\mathrm{SQ}}=5\hat{\mathrm{i}}-5\hat{\mathrm{j}}$

Area $=\frac{1}{2}\left|\overrightarrow{\mathrm{d}}_1 \times \overrightarrow{\mathrm{d}}_2\right|$



$=\frac{1}{2}\begin{vmatrix}\hat{\mathrm{i}} & \hat{\mathrm{j}} & \hat{\mathrm{k}} \\ 1 & 5 & 0 \\ 5 & -5 & 0\end{vmatrix}=\frac{1}{2}|(-5-25)\hat{\mathrm{k}}|=15$ **(square units)**

Q.72 Answer:

—

Solution:

We know that $\frac{\mathrm{dT}}{\mathrm{dt}}=-\mathrm{k}(\mathrm{T}-\mathrm{Tm})$, where

k is proportionality constant, $\mathrm{T}=$ Temperature of body, $\mathrm{Tm}=$ Temperature of surrounding.

$\therefore \frac{dT}{dt} = -k(T - 290) \Rightarrow \int \frac{dT}{T - 290} = \int -kdt$

$\log |T - 290| = -kt + C$

Initially, $t = 0,\ T = 370$

$\therefore \log |370 - 290| = 0 + C \Rightarrow C = \log |80|$

$\therefore \log |T - 290| = -kt + \log |80|$

When $t = 10,\ T = 330$

$\therefore \log |330 - 290| = -10k + \log |80|$

$\therefore \log \left[\left[\frac{40}{80}\right]\right] = -10k \Rightarrow k = \frac{1}{10} \log 2$

$\therefore \log[|T - 290|] = -\frac{\log 2}{10} t + \log |80|$

When $T = 295$, we write

$\log[295 - 290]] = -\frac{\log 2}{10} t + \log 80$

$\therefore \frac{(\log 5 - \log 80)}{\log 2} \backslash (-10) = 1$

$\frac{\log\left(\frac{1}{10}\right)}{\log 2} \backslash (-10) = 1$

$\therefore t = \frac{-4 \log 2}{\log 2} \backslash (-10)$

$= 40$

Q.73 Answer:

—

Solution:

$\gamma\gamma\gamma = \gamma(111)$

$= \gamma.3.37 = (\alpha\beta)(\gamma\beta)$

Clearly, γ is equal to the last digit of β^2. Now 37 must divide one of the $\alpha\beta$ or $\gamma\beta$ Let it be $\alpha\beta$ then $\alpha\beta =$ 37 or 74 (if $\gamma = 8$, then $\gamma.3.37 = 12.74$)

$\alpha\beta = 74$ is not possible because

$$\left.\begin{array}{c}(\alpha\beta)(\gamma\beta) \geq 74.14 = 1036 \\ \text{But } \gamma(111) = 12.74\end{array}\right\} \text{ for } \gamma = 8$$

$\therefore \alpha\beta = 37 \Rightarrow 5\beta = 27 \Rightarrow \beta = 7, \alpha = 3, \delta = 2 \ \& \ \gamma = 9$

$\therefore \alpha + \beta + \gamma + \varepsilon = 21$

Q.74 Answer:

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Solution:

$$y = \left(\sin^{-1} x\right)^2 + \left(\cos^{-1} x\right)^2$$

$$\therefore \quad \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{2\sin^{-1} x}{\sqrt{1-x^2}} - \frac{2\cos^{-1} x}{\sqrt{1-x^2}}$$

$$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{2\left(\sin^{-1} x - \cos^{-1} x\right)}{\sqrt{1-x^2}}$$

$$\Rightarrow \sqrt{1-x^2}\frac{\mathrm{d}y}{\mathrm{d}x} = 2\left(\sin^{-1} x - \cos^{-1} x\right)$$

Differentiating both sides w.r.t. x, we get

$$\sqrt{1-x^2} \cdot \frac{\mathrm{d}^2 y}{\mathrm{d}x^2} + \frac{\mathrm{d}y}{\mathrm{d}x} \cdot \frac{1}{2\sqrt{1-x^2}} \cdot (-2x)$$

$$= 2\left(\frac{1/\mathrm{AP}}{\sqrt{1-x^2}} - \frac{(-1)}{\sqrt{1-x^2}}\right) = \frac{4}{\sqrt{1-x^2}}$$

$$\therefore \left(1-x^2\right)\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} - x\frac{\mathrm{d}y}{\mathrm{d}x} = 4$$

Q.75 Answer:

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Solution:

Let $n(B) = x n(A) = x + 2$

$n(F) = y \qquad n(C) = y + 3$

$n(D) = z \qquad n(E) = z + 5$

$x + y + z + x + 2 + y + 3 + z + 5 = 40$

$x + y + z = 15$ **but** $x \geq 1 . y \geq 1$ **and** $z \geq 1$

So total ways of distribution $\mathrm{si}^{14}\mathrm{C}_3 = 91$